

Character documentation: Patricia

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1 Document introduction

1.1 Reading efficiently

To allow you to save time, I made some effort to separate purely numerical data from written descriptions. It is put in tables at the beginning of each chapter – I tried to write it in such a way that if you're not interested in the numbers you can just scroll past them & still understand what I wrote.

The table of contents is hyperlinked so you can just click on a chapter to be moved there, and itself is also pretty informative. If you're fine with reading short segments but struggle to read longer things, a good approach is to just open the document & read the relevant bit *once it becomes relevant*.

There are also some pictures throughout the text, and seeing them will give you information a lot quicker than text ever could, so I'd suggest you just scroll through the document to have a glance at them.

1.2 Reassurance for people who aren't fond of strictly adhering to real-world science or using math in sci-fi

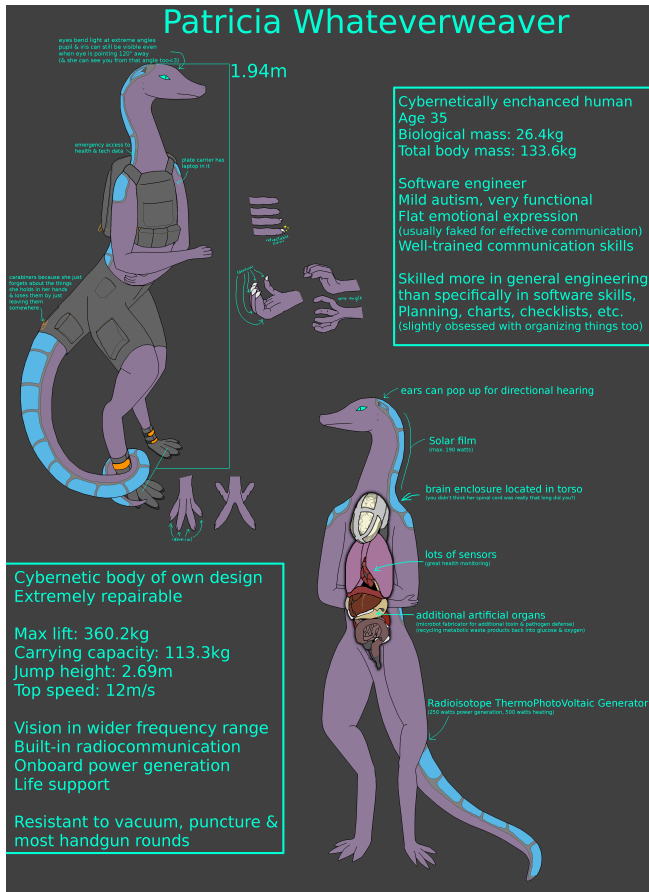
I noticed that hard-scifi fans don't have a particularly good reputation in roleplaying communities & I'd like to address that here.

I personally avoid the idea that realistic science fiction is somehow "better" than soft science fiction. I don't plan to push anyone to do math to check that their storytelling is realistic, or criticize people over using things that are completely made up.

I'm totally open to discussing changes I might have to make to my character for the sake of balance or consistence with the world in whatever setting I end up using this character in, as I realize the results of genuinely using math & science can produce things far more powerful than most people would be ready to address in their writing.

2 Basics & TL;DR

parameter	metric	imperial
mass	133.6kg	294.5lb
height	1.94m	6ft 4in



Physically: patricia is a human cyborg. Her current body is entirely of her own design, made exclusively of components that have publicly available schematics & had been in common use all around sol for centuries, making it fairly easy to repair in case of damage.

Mentally: she struggles with social contact since she has to consciously deal with a lot of things most people handle by instinct. She can act natural or even charming if she has to, but finds it tiring.

She still enjoys social contact under the right circumstances. Also has a compulsion to always plan things out & keep track of her things.

Her interests include transhumanism (obviously) & general engineering.

She also loves anything that deals with rough environmental conditions.

Her equipment: almost always the same set of clothes (custom boots, cargo shorts, plate carrier & backpack), a laptop in her plate carrier, an odd laser-projector-based smartphone-equivalent in her pocket & bunch of stuff in her backpack typically including a medkit & a small toolset for repairs.

Her origins: one of the countless rotating habitats in asteroid belt of the inner solar system. She was born on february 29th of 2400 & raised by the hard work of her parents rather than the calculated & carefully managed upbringing assisted by AI & child psychologists like most of her online friends had – she holds a grudge against the whole culture she grew up in for that.

She broke off from her family early into adulthood, and after saving enough to cover any unexpected problems followed her dream of a fully cybernetic body. Some time later she travelled to Ceres & got hired as engineer by an asteroid mining crew for a small share of the proffits, which made her moderately wealthy once they finally had a good find. The wealth & experience gained helped her later get certified for handling fissile material.

3 Biology

Biologically patricia is a fairly normal male human comparable to modern examples at the time of writing (2023) – mostly unchanged by evolutionary pressures of living in space & mostly untouched by radical genetic modification, save for minor improvements & genetic cherrypicking that ensured she be free of genetic defects, have perfectly well formed organs, low cancer risk, low chance of developing allergies, etc.

3.1 Nerve impulse conduction boost

Patricia has a modified version of a gene controlling the growth of nerves that results in thicker nerve fibers, including those of axons of neurons in her brain, which makes them transmit impulses much quicker & in turn gives her faster thought process & reaction time. Both are still within the same order of magnitude as that of a typical human, but it puts her reaction times far on the quick end of human norm. The effects of this are mainly just general convenience in all areas of life that require fast reactions & quick thinking, as well as slight inconvenience in form of further widening the gap between rate of thought & rate of speech.

3.2 Engineered gut flora

The bacteria in patricia's gut are of varieties intentionally modified to be more resistant to displacement by other bacteria & at the same time pose less of a threat if they were to ever get out into her abdomen or bloodstream as a result of physical damage to the intestinal tract. It also helps prevent intestinal blockage by enabling breakdown of cellulose & some common plastics, as a side effect also enabling her to extract more energy from food, or potentially even some things that usually aren't quite considered food.

They also bind various harmful compounds & undesirable elements like transuranics & toxic heavy metals into forms that will not be absorbed by her gut while also accelerating their removal. This effectively creates a minimum dose requirement for most common toxic substances to actually start building up in her body, at the relatively small cost of upset gut, which on the flip side could too be helpful as an early warning that a food or water source is contaminated.

3.3 Substance sensitivity (due to low biological body mass)

Due to her low biological mass she consumes a lot less food, and is also a lot more vulnerable to substances like caffeine or alcohol – 17ml of ethanol, the amount you'd usually find in a 340ml can of 5% beer, or a single shot glass of vodka, is enough for her to pass 0.1% BAC, which in 2023 is the highest legal limit for driving in any country on earth, and a level known to cause slurred speech.

Any kind of poisoning can be a lot more dangerous to her, and she usually requires lower but more frequent medication doses than most people when she takes any.

4 Personality & psychology

4.1 Sensory processing issues

Her brain does not filter stimuli quite as much, which can cause issues when information is not presented in clear form – she may have issues understanding words if while she’s talking to someone, other people nearby are also talking, and particularly noisy places can cause her actual physical discomfort, which is one reason she opts to communicate through text whenever possible. This is somewhat mitigated now that she has an artificial body that provides directional hearing & ability to pre-filter stimuli or outright turn them off.

The need for some peace & silence is a very intense one with her & can take priority over even basic physiological needs like warmth or hunger. If neglected it can actually have pretty severe psychological effects on her, from simple symptoms like irritation & inability to focus on the low to severe ones like visual distortions up to temporary blindness & loss of language comprehension.

4.2 Reduced affect

Patricia doesn't express emotions naturally. Her emotional display is typically done at extra effort just as a courtesy to others who use it naturally & rely on it for communication. She has to consciously think about her nonverbal communication for such things as letting people know that she's listening (which ironically distracts her from doing so).

She’s very skilled at social interaction & activities like smalltalk, and can be very charming if she wants to, but usually finds the act rather exhausting, which makes her a lot more comfortable in formal scenarios where the contact is rigid & structured as opposed to more casual ones.

4.3 Distraction sensitivity

People greatly distract patricia & she’s frequently frustrated that others don’t feel the same way. She has no issue with being alone & generally wouldn’t mind having no contact with anyone else for a day or two. When not getting distracted by others she can focus on a task extremely deeply & typically gets copious amounts of work done this way, though she also tends to get so stuck in that state that she even ignores her physiological needs.

4.4 Mild hypochondria

Patricia’s attitudes towards her artificial & biological parts are polar opposites. She tends to approach damage to her artificial body as mere annoyance or even be a little reckless with it knowing it can be repaired fairly easily. Any thought of damage to her biological organs though is very distressing & she’s often worried about germs, radiation exposure & food contamination, frequently checking herself for any deviations from norm. She’s also quite prone to placebo effect, so a mere belief that she is hurt can pretty easily cause her genuine pain, she might feel & behave like she’s sick after a situation where she could have gotten sick with something but actually didn’t & perfectly good food can give her food poisoning if she just thinks something was wrong with it.

4.5 Strict planning & methodicity (& distrust of own memory)

Patricia has a compulsion to always find the one correct, repeatable way to do something that she could stick with forever, without ever having to adjust, even if there isn't one right way to do it, and write it down in detail so that she does not have to rely on her memory.

She does not trust her own mind & doesn't see it as very reliable, preferring to offload anything she might forget to external storage, like that in hard drives & printed documents. She also just likes to have everything planned out ahead of time.

This shows frequently in her keeping various instructions & checklists for how to do certain things she does repeatedly, having a precise list of where all her things are that she updates whenever she moves anything & being very strict about doing things on time & in the right order. She'll often have a plan for what she wants to get done on a given day & will get upset if she has to adjust it even so much as just changing the time of when she does it, but otherwise doesn't adhere to a strict schedule.

It also applies to her social life & communication, where she likes to have everything structured in a consistent way – she has a tendency to introduce herself to new workplaces by handing someone what is effectively an instruction manual for herself, containing information about her skills, health, psychology & responses to various situations, including what tends to boost or degrade her performance & various less important likes & preferences, effectively detailing a complete social protocol for both herself & people interacting with her.

In more hidden ways she uses the same tactics to tend to a mild irrational fear of losing her memory, and keeps documents for her own viewing that detail such things as her own personality, everyday habits, what her past was & personal information like her own name, such that she could effectively regain all the information needed to be her normal self from them. Externally this only shows in occasional jokes referring to checklists & the like as better than her own mind.

4.6 Great motor neuroplasticity

Patricia's way of dealing with changes that require she rewires her muscle memory, such as from having to use new tools or most notably in her life having to learn how to use new body parts, tends to be of the brute force approach where she just goes out of her way to use it a lot till she develops the needed muscle memory. She is generally very insistent on that she should just be able to deal with the change & refuses to fall back on just using things she's already proficient with instead.

Since she had to adapt a lot to her new body, this approach actually resulted in her motor cortex getting very good at rewiring itself, and she can build or adapt her muscle memory much faster than an ordinary person.

5 Structural features

5.1 Skin

Her skin has several layers to it, they follow as such, starting from the outermost:

1. Cosmetic silicone hexagons secured by hot melt adhesive. They're flexible & can be easily removed with something like a soldering iron or a utility knife for replacement. Their purpose is to prevent skin wear from accumulating. Soles & palms are instead covered by larger pieces that imitate the surface of gecko paws.
2. Black [polychloroprene](#), incorporating a web of coolant lines, allowing it to be used as a low-temperature radiator for getting rid of heat. Over 60 times stronger than human skin, more electrically isolating & resistant to chemicals & temperatures up to 100°C. Made in one part, continuous across the whole body.
3. a thin layer of smooth teflon, meant to negate friction against the outer cover.
4. thin silicone incorporating a lot of pressure, temperature & continuity sensors.
5. Thin wire mesh – shielding from external electromagnetic interference & guard against leaking electromagnetic interference into the environment
6. 3mm layer of kevlar soaked in a sheer-thickening fluid & enclosed in silicone to prevent leakage of the fluid, [equivalent to 7.5mm of just kevlar](#) and providing protection on level with [VPAM standard PM 4 body armour](#) – capable of stopping nearly all handgun fire.
7. Repetition of layer 5
8. Repetition of layer 4.
9. Repetition of layer 3, for negating friction between skin & the internal components.

Thanks to having 2 sensing layers, she can easier tell severity of physical damage to her body & judge how quickly heat is moving through her skin.

The inclusion of kevlar is not actually such a big issue if she were to need medical attention. Contrary to popular belief, with a bit of work a utility knife would get through it just fine.

5.2 Skeleton

See chapter [15.4](#) for calculations.

Patricia's bones are made of a Titanium alloy containing 6% Aluminium & 4% Vanadium, typical for aerospace uses due to it's high strength & low density. They are also hollow, which uses the material more efficiently to minimize the weight while achieving high strength.

Their design is such that they will be able to handle twice the bending stress that she could possibly ever produce with her own muscles.

Every bone in her body is actually the same just with differing lengths, except for the spine & ribs. This does not have any practical benefits other than easier manufacturing & quality assurance.

5.3 Muscles

See chapter [15.3](#) for calculations & deeper details.

General muscle strength	201.4N/cm ²	Times stronger than human muscle	6.7
Max exerted force (1 arm)	903.5N	1 arm lift capacity in 1g	92kg
Max exerted vertical force (1 leg)	2422N	1 leg lift capacity in 1g	246.9kg
Max acceleration	2.23m/s ²	Top speed	12m/s
Total carrying capacity in 1g	113.3kg	Jump height in 1g	2.693m

Patricia's body uses Constrained Liquid Dielectric muscles, which are effectively powerful capacitors that use partially filled bags of fluid as insulator & crush themselves in one direction while expanding in the others when charged. Given enough current a muscle can contract to full force in just a few milliseconds, so reaction time is not an issue.

Every single muscle in Patricia's body is actually the same – she designed one standard muscle module which is linked together in series or parallel to form bigger muscles with various contraction factors & maximum forces depending on the configuration. This makes them easy to replace, and any standard fabricator capable of detail in the range of 0.1mm can be used to make those modules, with the only material that can't be 3d printed into the muscle being extra pure deionized water that acts as the liquid insulator in the muscle.

The charge used is mostly just circulated between the muscles, allowing the inefficient main battery to be skipped & letting her operate at extreme efficiency. This also makes her muscles a small backup power storage which she could use if her main battery were to fail, and will prevent her body from shutting down abruptly if power is lost, like an additional 89 watt-hour battery.

Voltage within the muscles can change depending on external circumstances – if the muscle is being stretched while it has some charge in it, the voltage inside it will raise, either making it harder to push more charge into the muscle or raising the energy that can be pulled out of it as the charge is removed to relax it. Her system fully utilizes this quirk & she will actually gain electrical energy from such activities as walking downhill or moving something heavy to a lower position.

Her strength is superhuman for her size & mass, but not in absolute terms - a professional weight lifter would match her. Her potential speed however would be very difficult to match, as her muscle can deliver far more force for the same amount of weight.

5.4 Manual external manipulation ports

4 spots on Patricia's sides are tunnels that go all the way through the skin, sealed with plugs that provide the same protection as her skin but can be unlocked from the inside & pushed out.

Once opened, the spot is still covered by a non-stretchable material that forms a sleeve which could be used by one of Patricia's fleshy tentacles (see anatomy section) to manipulate or at least feel objects outside the body without relying on any electronic systems.

6 Environmental adaptation

6.1 Heat management system

Throughout patricia's artificial body runs a net of coolant lines, separated into multiple loops as to keep damage in one spot from being able to drain coolant from the entire system. Efficient heatpumps are used to move the heat from the inner loops to outer ones running just under her skin, that transfer the heat away into the environment.

Since the heatpumps work both ways, she can actually absorb heat from the environment if that's needed. For cases when that doesn't work there's also simple resistive heating built into some of the inner coolant loops, but usually the heat from her RTPVG is just redirected to be used instead.

In case of a potentially fatal health emergency it will be used to rapidly cool her organs & keep them a few degrees above freezing to maximize the length of time in which she could be recovered & revived without sustaining damage to her brain.

If the heat load on her body is somehow simply too much to be carried away by her skin, it's actually capable of pulling water from the reservoirs in her neck (chapter 6.2) to be used for non-regenerative cooling. The water would be pumped via ducts to her RCS nozzles (chapter 6.5), which get heated up via heatpumps, cooling the rest of the system & vaporizing the water. This actually provides [more thrust per kilogram spent](#) than just air like she usually uses in her RCS system, though is generally not useable due to how nasty jets of steam can be to things around her. Every gram of water vaporized takes 2257 joules of heat away from her body, though heat created by the running heatpumps allowing this to happen makes that slightly less in practice, but in theory 750ml of water would let her get rid of nearly 1.7 megajoules of heat very quickly, and much more if she can readily refill her water stores.

6.2 Breath temperature & humidity adjustment

Max. water collection	250ml/h	Max. power use	750w
Max water stored	750ml		

Air passing through her nose has it's temperature & humidity adjusted by systems typical for standard resistive heaters, heatpumps & evaporative humidifiers to help her endure extreme temperatures & ensure the air she's breathing isn't too dry or to help conserve water by pulling moisture from the air she exhales. It also means her breath does not significantly deposit moisture anywhere such as on a window, piece of electronic she's looking closely at, or the visor of a spacesuit. The collected water is stored in her tongue & some reservoirs in her neck. If it fills up too much she'll just have to release some of the water through a hole at the tip of her tongue.

Under normal circumstances this system will not have to run much, but she does have a habit of keeping it active & set for water collection a lot for simple convenience – in livable environments it can actually collect enough water to completely satisfy her biological need for water intake.

6.3 Space hardening

Patricia's body will immediately seal all its orifices when it detects sufficiently low pressure or a quick drop in pressure. This combines well with her skin not being stretchy enough to let her lungs or other organs expand to a dangerous degree when external pressure is removed.

She is not susceptible to barotrauma from sudden loss of atmosphere & in most scenarios could safely remain in vacuum indefinitely, provided she has power for her metabolic recycler.

As one of the measures meant to potentially let her operate in space unprotected, the dye in her skin is actually UV-reactive, and turns pearly white when exposed to concentrated hard UV light like you would find in unfiltered sunlight, or see used in hospitals to sterilize rooms. This is so that she can reflect more light, otherwise she could overheat very easily in vacuum. To a lesser extent it also responds to soft UV which may be good fun if you have a blacklight.

6.4 Radiation detection

Several radiation sensors are located in Patricia's body in her torso & head. They let her detect dangerous radiation & keep track of the exact dose she'd been exposed to, as well as triangulate the sources of the radiation to help her avoid them.

6.5 0-g Maneuvering System

Calculations for these numbers in chapter [15.2](#)

ISP (air)	65s
Stored impulse (air)	310Ns
Δv at 133.6kg (air)	2.32m/s
ISP (h2o)	150s
Stored impulse (h2o)	1103Ns

Stored air mass	0.486kg
Stored air volume	405L / 0.405m ³
storage pressure	450 bar
Stored water mass	750 gram
Δv at 133.6kg (h2o)	8.25m/s

There are several cold gas thrusters on her shoulders & on the middle of her tail on its sides, which gives her 0g maneuverability. The gas used by them is simply air, usually collected at any opportunity, stored pressurized in an air tank in her tail.

It's mainly meant to be used in already pressurized environments, but could potentially be used to maneuver in vacuum, or to repressurize a spacesuit or any other small space with air at atmospheric pressure.

The nozzles can be heated using heatpumps, and then vaporize water as fuel, which serves as a method of cooling more than propulsion, though it does provide superior results to using pressurized air at the cost of being potentially dangerous to any people or equipment nearby.

7 Energy management

7.1 Aluminium-ion-germanium-nanowire batteries

Battery energy density	2.4kWh/kg
Total battery mass:	18kg / 39.68lb
Total energy stored:	43.2kWh / 155.5MJ

Her body uses mainly [Aluminium-Ion batteries](#) for storing electricity, which instead of the standard graphite cathode use [germanium nanowire](#), increasing their capacity by 2.4 times & reducing the rate at which the batteries lose capacity from being charged & discharged to just 0.01% per cycle.

7.2 solar film

Solar film area	0.45m ²	Solar-electric conversion efficiency	47%
Geometric efficiency	67%	Max. output in earthly sunlight (1361w/m ²)	190w

The back of patricia's neck, torso & tail are covered in a highly efficient solar film. It's more of a style element, she doesn't sustain herself entirely on it's power output under normal circumstances, but in the event she were to get spaced or stranded somewhere it could help her stay alive. Even though it's only meant for emergencies, the earth-sunlight-mimicking internal lighting of most space habitats is enough for her to achieve net negative power consumption in moments of low activity.

7.3 muscle cell

Peak power	500w
Runtime till exhaustion	around 600s

Among patricia's organic internals, there is one gloriously huge muscle that had been connected to power-generating systems as yet another emergency power generation feature, and as a means of allowing her to exercise despite the lack of biological limbs.

7.4 radioisotope thermophotovoltaic generator

See chapter [15.1](#) for calculations

Power	250w	Fuel mass	2.88kg
efficiency	50%	Heat generation	500w

Located near the end of her tail is an RTG utilizing [thermophotovoltaic cells](#) to boost it's efficiency. It uses standard [General Purpose Heat Sources](#) & usually uses the surface of the tail as the radiating surface, making it slightly warm to touch, though this heat can also be redirected anywhere else it's needed. For safety reasons in case of power loss, some transfer of heat from the RTPVG to the surface of the tail is accomplished with simple flexible heatpipes that don't need a pump to work.

8 Health features

8.1 Metabolic recycler

Oxygen recycling efficiency	32%	Peak power use	120w
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Patricia's bloodstream is connected to a metabolic recycler, able to conduct chemical reactions to correct for off-nominal concentrations of some simple chemicals like oxygen, carbon dioxide, glucose & most common vitamins. It can do nothing to correct levels of more complex chemicals like proteins though, and neither can it do nuclear transmutation to correct for deficiencies of elements. It's also connected to her kidneys for the purpose of recycling urine in case she does not have access to fresh water & needs to conserve it, or is not able to urinate as would be the case if she were spaced without a spacesuit.

It also provides data about concentrations of various chemicals in her blood, since it has to run the tests anyway to function properly, which among other things is also processed to give instructions to the bloodstream-integrated microfabricator.

8.2 Diagnostic suite

Most mechanical components in patricia's body are being frequently checked for signs of wear or malfunction, her internal organs are frequently scanned for any abnormalities & blood tested for levels of various biomarkers & chemicals. All the information is collected & stored to be available to her for review at any time, and she will be alerted upon any significant findings.

Upon passing certain thresholds, red lights on the sides of her neck will begin flashing to bring the attention of anyone nearby to a pullable cover labeled "diagnostic suite" with a port & button to be held down to activate external access, that then offers the interface to any connected device in form of a web page, containing all the readings of her internal sensors, blood test results, component test results, various analysis of the data by the diagnostic programs as well as her entire medical history, the entire blueprint of her body & schematics of every component used in it in a standard format accepted by most engineering software & automatic fabricators.

If the visual signals are not acted upon for a set time or condition worsens enough, it will periodically pass a message to her radiocommunication systems to be emitted on the standard distress frequency.

8.3 Bloodstream-integrated microfabricator

A machine is integrated into patricia's bloodstream, which is capable of assembling biodegradable microscale mechanisms & producing complex nanoscopic particles from given blueprints. It uses solutions that may sometimes have to be refilled, but it's also fitted with components that filters excesses of various elements out of her bloodstream as an alternative way of replacing them, which can be enough to never worry about refilling it manually when the use is low.

It can be used to fight pathogens her actual immune system doesn't protect her against yet, by manufacturing microscopic machines specifically made to fight it off from a selected schematic, or it could make ones that increase blood oxygen carrying capacity or soak in a particular toxin (like the alcohol mentioned in biology section) – that is, *if* such a schematic is available. She does carry sets of schematics helpful against most toxins & diseases, and it responds to readings from the metabolic recycler to remove any undesirables, but it might not help much against previously unknown poisons or diseases.

The machines it creates are made mostly of things that can be naturally found in the human body, so they're not the rigid metallic kind. It also does not manufacture every single one directly, and instead creates machines that create smaller machines that create smaller machines & so on, with the last in this line of tiny factories producing the desired product. This is done in a fluid-filled sack that it filters the end product from later & allows it to avoid a lot of the wear & tear it would experience if it had to do all the work directly. Safety in that process is ensured by incorporating [selenium](#), [chromium](#) & [molybdenum](#) in critical parts of the microscopic factories it makes, in a way that makes it impossible to simply use some other elements in their place without completely re-engineering the microbots – all 3 are present in the human body at extremely low concentrations, far too low to sustain continued function of the microfactories, but are present at high concentrations in the liquid filling it's manufacturing chamber. The actual amounts are still low enough that even accidental spillage into bloodstream by rupture should not cause toxicity.

8.4 Artificial heart

Patricia's organic heart had not been replaced, it is still there, but an artificial one is also connected to her bloodstream in parallel. If her heart were to stop or get stuck in a rhythm that cannot pump blood properly, it would soon take over to maintain the circulation.

9 Information processing & communication

9.1 Data storage

Patricia's body contains a lot of data storage, having 16 petabytes capacity in form of [Superman Memory Crystals](#) (yes they are read-write capable), as well as 100 terabytes of cache of standard flash memory due to SMC having very slow write speeds in comparison.

Among other things, she has the entirety of wikipedia downloaded onto it & full archives of learning courses for every field of science & engineering from several popular learning sites, as well as every single video made by a few of her favourite online content creators.

9.2 Radiocommunication suite

Patricia's body incorporates variable antennas (needs link) that can tune to any frequency in the ranges from VLF to THF (frequencies 3kHz to 3THz), allowing her to receive signal from all radio-based positioning systems & receive all common radio communication, in case of unencrypted signals even those that are not meant for her. For this reason, law related to eavesdropping on unencrypted radiocommunication is one of the things she looks up before visiting places.

She only has the ability to transmit in the range between THF & HF (3THz to 3MHz) due to antenna size constraints, but usually keeps the variable transmitters completely disabled & only uses ones specifically designed for operating on some select open frequencies as to avoid accidentally interfering with important radio signals or just emitting stuff on frequencies she does not have required licenses for.

Among all that is included a particularly powerful transmitter purpose-made for transmitting on the standard distress frequency.

9.3 Near Field Communication module

An antenna coil located in Patricia's snout allows her to use NFC services – among other things:

- reading data from electronic tags
- emulating smart cards & key tags for opening electronic locks
- sending/receiving files to/from NFC-enabled devices

The primary use of it is to open electronic locks - having the feature built into her body as opposed to being an external electronic device greatly increases security, since it means she will never lose it & lock herself out or have a key stolen by someone seeking unauthorized access.

9.4 Standard computing hardware & operating system (write more)

Patricia's body runs a full operating system within it, providing her a virtual reality interface with the ability to configure certain components in her body & check the data they provide as well as run various apps.

It's a [distributed system](#) running on multiple processors with their own separate memories, rather inefficient as it prioritizes robustness over efficient use of computational resources, even to such a degree that the memory & processors could be hot swapped without much chance of causing damage or major alteration to the system & software running on it – there is no failure mode where a single component failure, much less a bit flip, would cause it to crash or corrupt it's contents & it could even continue to function in a case where [up to 25%](#) of the processors in her body begin maliciously malfunctioning against the rest, either from damage or tampering.

She has several applications installed on it, but keeps it rather clean compared to her personal electronic devices & stays away from impulse-installing applications or even just connecting to the internet for the sake of security.

9.5 Status lights

The gray areas on her neck, torso & tail, which serve as the border between her body & the solar pannel section, are fitted with LED strips, which allows them to light up in any colour or combination of colours.

They mostly serve to signal problems to the outside – major problems will have them light up in red, repeatedly fading in & out, bringing attention of people nearby to the fact that something isn't right. Other blinking patterns signal different kinds and severities of problems.

When not used for it's purpose of signaling problems it tends to fall to more silly uses like just displaying a rainbow RGB pattern or other cosmetic uses.

10 Backup systems

10.1 Alternate control computer

Patricia's body has a redundant set of controls meant to take over in the case that the distributed computer system that usually controls it becomes faulty beyond recovery. The backup systems are made of simpler, more robust electronic circuits with few to none customization options.

If the main system fails, a set of 3 identical circuits running more power-hungry but far more interference-resilient transistor-transistor logic takes over & provides patricia with control of her own body according to pre-programmed functions. The control is mostly the same, except for lacking any advanced functions – her senses revert to what is equivalent to standard human senses, and the health-critical systems are switched to only rely on one another & their own internal controls instead of the main system.

10.2 Manual controls

Inside of the cavity that holds the container of patricia's organs are manual controls that can be used to cut power to various parts of the body or reset specific systems. They're all carefully positioned & take a fair bit of force being used to *pull* on them, so that she would not accidentally activate any of them, and are all labeled in braille as otherwise she would have no way to tell what she's doing in the event she actually has to use those.

11 Sensory equipment

11.1 Hearing capability

Min. hearing threshold	-23dB / 12.5×10^{-6} Pa	Frequency range	0.1hz-100khz
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Her ears can detect noises at the level of the ticking of a mechanical wristwatch 3 meters away. She can adjust her threshold of hearing to anything above that though & frequently does so, as having it at the lowest is certain to be annoying in most settings. The intensity of signal that is actually passed to her auditory nerves is also capped to a level set by her as to never exceed a comfortable threshold.

The range of frequencies she can hear is typically what is standard for a human: 20hz-20khz, with a smooth dropoff in sensitivity at the ends. She can set any sensitivity curve she wants into it & apply various stretching & offsetting to the sound before it's passed to her, so if she wishes to she could actually hear at frequencies used by seismic or ultrasonic data transmission & certain sensors.

When directional hearing is needed, the film near her ears can pop up to form a standing ear that focuses sound coming from the front onto the actual microphone. Otherwise, they appear completely flat, nothing more than a square of slightly textured surface.

Snout can also detect surface vibrations & interpret them as sound for communication in vacuum.

11.2 Advanced optics

Patricia's eyes are an exception to the overall ease of repairability of her body. They are still her own design, but each eyeball is a single very complex component, which requires nanoscale manufacturing processes to create. It's difficult to damage, but once it is damaged it would be so hard to fix that simply making a whole new one is genuinely less costly in time, effort & funds.

The eyes can be removed from their sockets for replacement fairly easily – the eyeball simply has to be pulled on with enough force to pop out, typically with some kind of suction tool, and a new one can be pushed in. The back of the eyeball is equipped with a standard androgynous usb port, and so is the inside of patricia's eye socket.

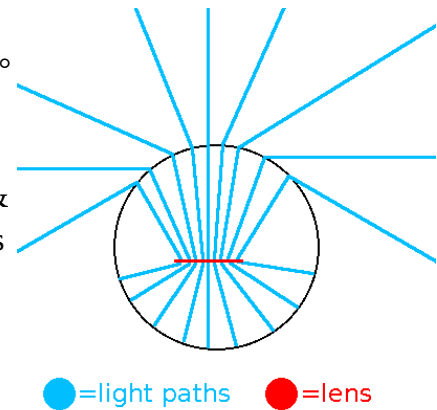
Exposure is adjusted by artificial iris that uses [ionic metal-polymer composites](#), sidestepping the common issue of shock-sensitive camera shutters. It can take on any shape & usually forms a vertical slit, but in darkness can dialate all the way to a circle with 3cm diameter. The same method is used for handling any moving parts inside the eyeball, as the soft mechanisms are far more resilient to damage from excessive acceleration.

The eyeball actually works like multiple webcams connected to a usb hub – it offers monochrome feeds for each of the 6 colour channels it receives & can be given instructions for adjusting zoom & exposure. 6 monochrome channels is not very useful alone, so they have to be mixed together in the desired way & converted into nerve impulses by other software & hardware.



11.2.1 Negative refractive index lensing

The eyeball lenses light with the help of materials that have [negative refractive index](#), letting the light be bent by more than 90° when it passes into the eyeball, which allows light from a very wide range of angles to be redirected into the eyeball. A set of moving flat lenses inside the eyeball allow for adjusting contrast & field of vision, effectively letting her zoom in on objects she wants to see better – on minimum zoom her field of vision reaches 240°, though in such extreme use [chromatic aberration](#) starts becoming problematic. Nonetheless, she often uses it just for the additional situational awareness.



Normally her eyeballs only redirect light from a 170° cone onto the sensor, but that can be reduced all the way down to 20°, effectively offering an 8 times zoom over normal human vision.

The extreme bending of light gives her eyes an odd appearance when viewed from very high angles, as the iris & pupil will appear compressed but still be visible even when the eye is pointing more than 90 degrees away from the viewer.

(remind me to draw a better, more accurate explanation picture)

11.2.2 Spectral sight

Patricia's eyes provide multispectral vision, allowing her to see beyond the human visible range.

They can detect light at 6 different frequency ranges:

- UV-A / soft UV (315nm–400nm) – not absorbed by atmosphere & visible to many of terran insects & birds
- Red (625nm–740nm), Green (495nm–570nm), Blue (450nm–495nm)– human standard
- IR-A / Near-IR (750nm–1400nm) – used by most night vision systems & occasionally for communication between electronic devices. Thermal emissions of objects hotter than 1800°C also start to peak in this range &
- IR-C / Thermal IR (8000nm–15000nm) – thermal imaging, emissions of objects with temperature between the freezing & boiling point of water peak in this range

She tends to stay within the human standard colour vision, but also spends a lot of time using a preset that reuses some of the nerves which previously took signal from red & green receptors for UV-A & IR-A to let her see in 5 colours. Her brain is by now somewhat used to that & she can tell each colour apart, but still somewhat associates UV-A with red & IR-A with green, rather than processing them as completely separate colours.

When going out to places where she might encounter devices using the IR-A range for communication she simply switches back to human standard preset, as seeing the actual infrared signaling can be quite dazzling.

11.3 Optical smell & taste sensors

Unlike a natural human nose which uses chemical interactions to detect compounds in the air, Patricia's uses lasers & optical sensors. A series of laser emitters shine light of very specific wavelengths through the air & measure the absorption of light on the other end, as well as scattering of light from the beam & any emissions from compounds that may have been excited by the laser light.

The data is processed & nerve signals are created to represent it, based on compounds most humans would be able to smell, with small modifications like assigning irritating smells to compounds that are dangerous but aren't detectable by natural human smell.

The processing can be configured very flexibly, for example assigning a strong smell to some very specific chemical compound, but the data from the sensors is also available raw for her operating system, allowing custom programmed responses like closing airways to prevent poisoning.

Thanks to additional nostrills, she can actually pull air *through* the sensors rather than repeatedly into & out of them, greatly increasing the sensitivity of her smell.

Due to the optical nature of the sensors, she is particularly sensitive to large particulate matter in the air such as soot or dust, and even foggy air will cause some sensation due to the light it disperses.

Her tongue works in much the same way – it's filled with water & has a lot of small pores, through which the water-soluble chemicals from food will be sucked in or just diffuse in through, and be subjected to the same absorption-emission-scattering analysis.

11.4 Additional notes

If you're here for what she can *detect*, here are some more things:

Chapter 6.4 - radiation detection.

12 Anatomy

12.1 Internal organs

organ	mass
intestines	3.5kg
liver	1.5kg
brain	1.4kg
lungs	1kg
stomach	0.9kg
heart	0.33kg
kidneys	0.3kg
spleen	0.14kg
pancreas	0.08kg
gallbladder	0.05kg
thymus	0.045kg
testes	0.03kg
thyroid	0.02kg

tissue	Total mass
skeletal muscle	5kg
Fat tissue	5kg
bone	1kg
Bone marrow	3kg
blood	3kg

Total biomass	26.4kg
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All the organs mentioned above are located in the torso, kept in place with artificial connective tissue that supports them in any position relative to direction of gravity, unlike that in a normal human body that is designed to spend majority time in upright position.

They are arranged roughly in the layout that would be expected of human organs (with exception of the brain, which as mentioned, is located in the torso) & contained in an additional protective bag, which, should there ever be a need, could actually be completely detached from the rest of her artificial body & removed from it with all her organs still inside, and be connected to a different system or a new body.

There are some artificial additions, in form of sensors stuck onto various organs monitoring their health & two major artificial organs – the bloodstream-integrated microfabricator & the metabolic recycler.



12.2 Marrow bone

There's a semi-artificial but completely biological organ in Patricia's body. A shell of bone, holding a lot of bone marrow, located behind her liver & connected to the major blood vessels leaving it. It contains all the bone marrow from the removed parts of her body so that she would continue to produce red blood cells.

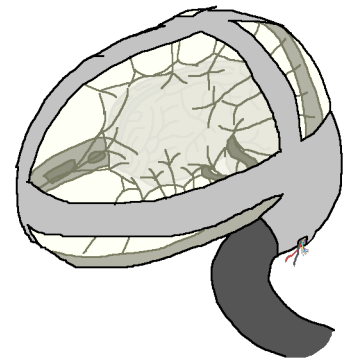
12.3 Biological Tentacles

Another semi-artificial but completely biological organ (or rather 4 of them). Patricia's own cloned skeletal muscle had been grown into 4 tentacles, each roughly 60cm long, which are connected to her nervous system & bloodstream. They usually sit in 4 sleeves formed by the walls of the organ bag & can be used to operate physical controls located inside the body, as well as in absolute necessity possibly reach outside the body.

12.4 Brain enclosure

Her brain resides in an artificial enclosure quite a bit bigger than a natural skull, which is located within the top of her torso, roughly on level with her shoulders. This position means smaller height difference between brain & heart, letting her handle higher G-forces.

The brain is covered in its natural meninges, which is then covered with a soft layer connected to the enclosure by elastic strings to gently keeping the brain in the middle of the enclosure, ensuring a good layer of fluid on every side of it that will further dampen any impact, preventing damage even by g-forces in the low hundreds.



The enclosure is mostly made of [aluminium oxynitride](#) to allow visual inspection & scanning in case of emergency, so that the enclosure would not have to be needlessly opened & risk whatever chance of contamination or infection there might be. It's strong enough to resist high-caliber gunfire or even explosions & has a layer of transparent bio-safe polymer coating the internal walls provide additional safety should the enclosure somehow be cracked. It's also equipped with sensors of its own attached to the inside to monitor the brain & enclosure itself for damage or any abnormalities.

The cranial nerves are connected to nerve interfaces within the enclosure while the spinal cord exits & continues out of it, wrapped in an artificial spine replacement, connecting to the rest of her natural organs like it would in a normal human body.

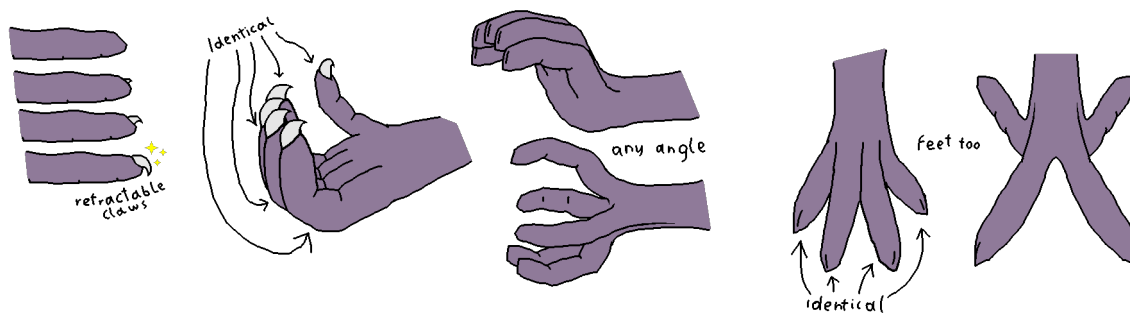
In case that her circulation ceases it can release a large dose of [glutamate receptor blockers](#) to prevent her neurons from [firing excessively & undergoing apoptosis](#) due to hypoxia.

12.5 Hands & feet

Each of her fingers & toes has a single sharp claw in it, which can be extended or retracted as needed. They are hook-shaped & mainly meant for climbing use, and are only sharp at the tip, rather than all along the edge, making them not very capable of cutting anything (whether intentionally or by accident). The material they're made of is nitinol, which lets them retain the sharp tip much better.

Her palms & soles are covered in microscopic structures that allow for adhesion to very flat & smooth surfaces via the Van der Waals force, also useful for climbing.

Her fingers can come apart so far that her middle finger would be pointing the complete opposite direction to her thumb & pinky, or even become opposing to all the other 4 fingers – roughly similar thing happens with her toes, and they can grab just as well. Also quite useful for climbing.



There is actually no anatomical sign to tell which hand/foot is the right or the left one, the right half of the hand is symmetrical to the left half, same with the foot – on both sides of the body, the hands & feet are just exactly identical.

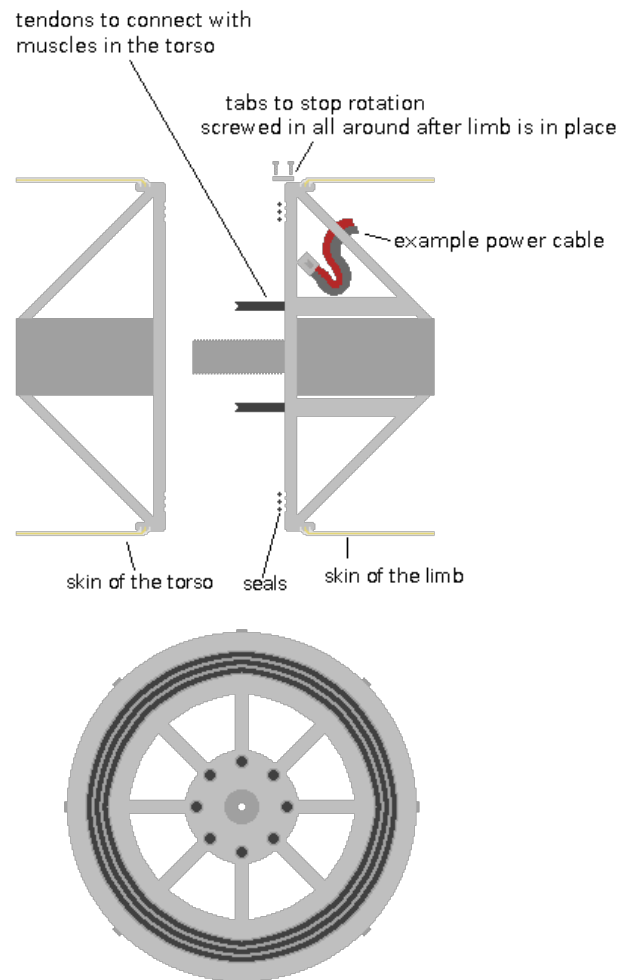
12.6 Limbs & body modularity.

Patricia's limbs are each their own whole integrated part, but they are not together one whole. The torso, two arms, two legs, tail & the neck+head combo, are each their own module. Each have a set of computing nodes built into them to manage their power use & provide a simple interface for control.

Each arm & leg is actually identical – the arm on the left side of her body & the arm on the right side are not even mirrored, they are simply exact replicas of one another, with the shoulder bent in the opposite direction - the shoulder joint has a full 180° range of abduction motion. The legs are in a similar situation, but while the hip joints have the same 180° range of motion, the way the muscles of the torso interact with the joint does not allow anything more than a full split.

The points where the limbs connect to the torso are designed to be possible to separate nondestructively, though not necessarily easily. Their power & data lines are joined with connectors & the skin is separated except for the outermost layer. The skeleton is joined between the torso & the limbs by the limbs simply having an extended threaded part on the bone, which screws into the corresponding slot in the torso, and then secured in place by screwing in additional tabs into the limb to keep the connection from rotating apart.

Pretty much none of the limbs show any indication of which side is the right side & which is the left side – they are symmetrical just like the hands & feet. While this does not matter for the tail & head, it allows using the same replacement arm or leg module to replace *either* the left or right limb.



12.7 Head & neck (+ vocal tract)

The superman crystal data storage is located in her head, right where you'd expect the brain to be if it wasn't located in her torso. It makes use of her natural instinct to protect her head from getting hit, as it is relatively more delicate than most of the machinery in her body. That's however not where the SSD cache is located, since sensory recordings would probably be critical to preserve following any scenario that sees her losing her head.

Her mouth is full of teeth which are all roughly conical, 6cm long & 2cm in diameter at the base, flattened on the outer side & gently curved so the tips are pointing 5 degrees inwards when extended. They are indeed sharp, but the length is picked carefully so that they cannot be legally considered a weapon.

They're hinged & become completely horizontal when she's not using them, or when she has to chew something hard, in which case their outer sides act as the crushing surfaces. The material used to make them is nitinol, just like the claws, and for the same benefits, plus one more, that of nitinol being completely biologically safe, so that it will not possibly react with any food consumed.

The vocal tract imitates that of a human, but differs enough for significance. The large distance between the lips & the start of the throat make all the plosives made at the back of the throat sound more guttural, as well as allowing her to access some of the forbidden fields of the phonetic alphabet. The shape of her mouth also creates occasional issues in pronouncing the letter r & other rhotics, resulting in her sometimes omitting them such as by mispronouncing there's as thess.

At the back of her throat, under a fluid-proof layer, there are small speakers, to be used as a backup if for any reason she cannot use her vocal tract – be it because it is sealed due to detection of a sudden pressure drop or toxic gasses, or that she cannot breathe for some other reasons. It does not sound as good as her usual way of vocalization though, as the speakers cannot produce the full spectrum of frequencies, especially the lower ones, not significantly affecting the actual frequency range in which she can vocalize, but making the sound not quite as rich, with the fluid-proofing also adding a slight muffled quality to it.

Her snout actually includes a third unusual way of vocalization, having a built-in contact speaker, and can pass sound vibrations into any solid material it is pressed against, potentially allowing communication in vacuum when radiocommunication is impossible or undesirable.

It can also act as a contact microphone, collecting vibrations from whatever surface it's pressed to.

12.8 Tail

It contains 18 kilograms of Aluminium-Ion batteries, a small pressure tank for air storage & a small radioisotope thermophotovoltaic generator. There are ports for charging & interfacing with external electronics around its base, as well as one hidden under a cover at the very tip.

Its shape can be roughly approximated by a cylinder from base to the middle, and a cone from middle to tip, though in reality the transition is a lot smoother.

Her design constraint of keeping her body water-dense to feel more fleshlike results in particularly much empty space here due to high density of the batteries.

13 Equipment & belongings

13.1 Standard clothing

Patricia's standard outfit consists of custom velcro-closed boots & cargo pants as well as a standard backpack & a plate carrier, all coloured in dark gray with bright orange details. She doesn't actually have an armour plate to put in her carrier & instead uses it for her laptop. There's usually a carabiner attached to the back of her backpack & to the back of her pants on the tail.

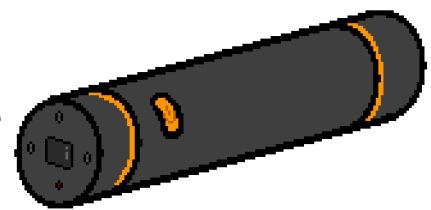
Due to how the clothing fits her body, she sometimes jokingly refers to it as "tactical midriff".

13.2 Personal (external) electronics

13.2.1 Laser phone

A phone in very nonstandard form factor. Instead of a typical display it uses a laser projector located at the front – it's far easier to find & replace a laser diode, a motor or a mirror than it is to replace an LCD screen.

It can still use an external screen if needed though.



Both front & back screw in place with rubber seals for replacement or access to the insides. The charging port is at the back & is closed by a rubber cover. So long as closed it can indefinitely resist vacuum, dust & immersion in shallow water.

It supports all the usual functions of a typical smartphone, though as a bonus it can communicate with similar devices via infrared laser, with very high transfer speeds over very long distances at the cost of needing line of sight & having to be pointed in roughly the right direction.

She made it herself, though she did not design it. It's a free schematic available on the web, made exclusively out of commonly used or trivially manufacturable parts.

13.2.2 DIY rugged laptop

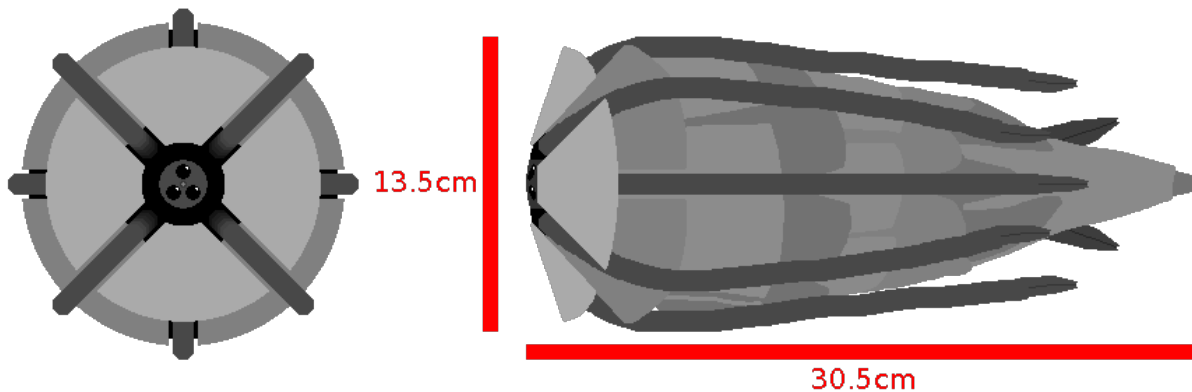
Just like the laser phone, it's a free schematic available on the internet which can be put together from common components & 3d-printed parts.

It's a heavy brick measuring 5×18×32cm with a completely waterproof rubbery cover & no visible air intakes for cooling. The back of it is covered in solar film that has a protective coating on it, slightly hurting it's efficiency but preventing it from getting damaged & allowing it to charge from sunlight.

Several switches on the side of the keyboard can be used to physically enable/disable touchscreen, geolocation & wireless networking. The keyboard can also light up to make the keys easier to see when using it in the dark.

It's usually stored in the ballistic plate slot of her plate carrier vest for convenient access.

13.3 Sairene – AI pet / personal assistant



Sairene is the result of Patricia attempting a DIY project she found online. It's a versatile personal assistant ran entirely by one of the commonly available large language models.

It's been tweaked a lot to be more conversable & feel a bit more like a person, but still has a tendency to remind people that it is just an AI assistant & all the various things like that it's advice should not replace proper planning or to not prioritize it over a person in case of emergency, listing potential negative effects of developing emotional attachment to an AI assistant & so on, though she now likes to point out that her name is NOT just a disguised phonetic clone of “siren”, and instead the ai is spelled like the letter a in the alphabet while the ene is like the een in keen.

All sensory input is given to sairene in text, and all of her actions are just interpretations of her text output. She can only see what the object recognition software tells her is in her vision unless she prints the raw picture data to herself & processes it with difficulty, moves mostly by tasking the navigation program with reaching some location, and has to output very specific commands to control any of the 8 tentacles for precise movement. There are also speakers & a set of antennas on the inside. The actual electronics are sealed in waterproof compartments.

Despite her crudeness, she is entirely capable of very broad problemsolving & complex reasoning, though not quite as well as a human, and thanks to the constant inflow of data can act on her own initiative with no input from a person.

From the bodily side, it's equipped with aluminium plates for protection (with edges safely dulled), 8 tentacles for object manipulation & transportation, a compressed air propulsion system for 0g movement powered from a common 5cm-internal-diameter spherical pressure tank, and a camera/microphone set at the front, with an included LED light. It masses about 2.7kg.

Sairene can sometimes be seen clinging onto patricia's shoulder or looking out of her backpack.

As with most things patricia makes, all the components in sairene are commonly used components designed a very long time ago, which stood the test of time in their reliability & versatility, such that they can now be very commonly acquired pretty much anywhere.

13.4 Custom mining suit

Patricia had to get a custom spacesuit due to her body shape, though this time she outsourced the design – nonetheless, just like most things she owns, it's charmingly low-tech & easily repairable.

The helmet has it's odd shape similar to the head of an axe due to the fact she needs quite a bit of space to open her mouth due to her snout.

It assists movement using hydraulics that provide many times more strength than patricia has, enabling fluid & easy movement despite the mass of the suit. Thanks to a powerful pump, the movement can be about as fast as patricia's usual movement.

It's designed primarily for mining purposes, providing great protection against crushing by inertia of high-mass objects, debris from undesired detonation of mining explosives, and any potential micrometeorite impacts.

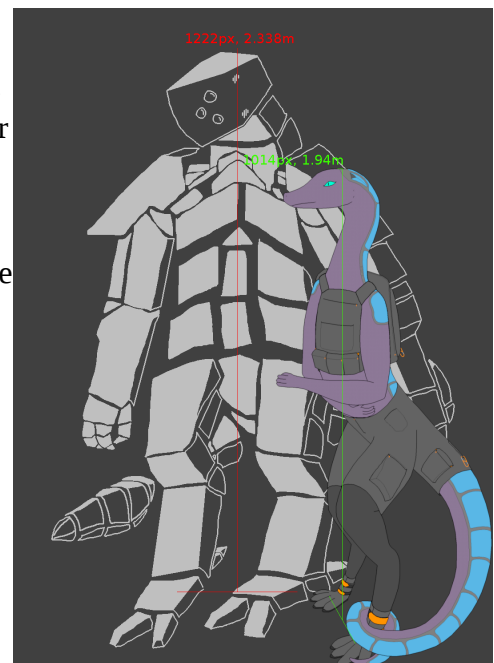
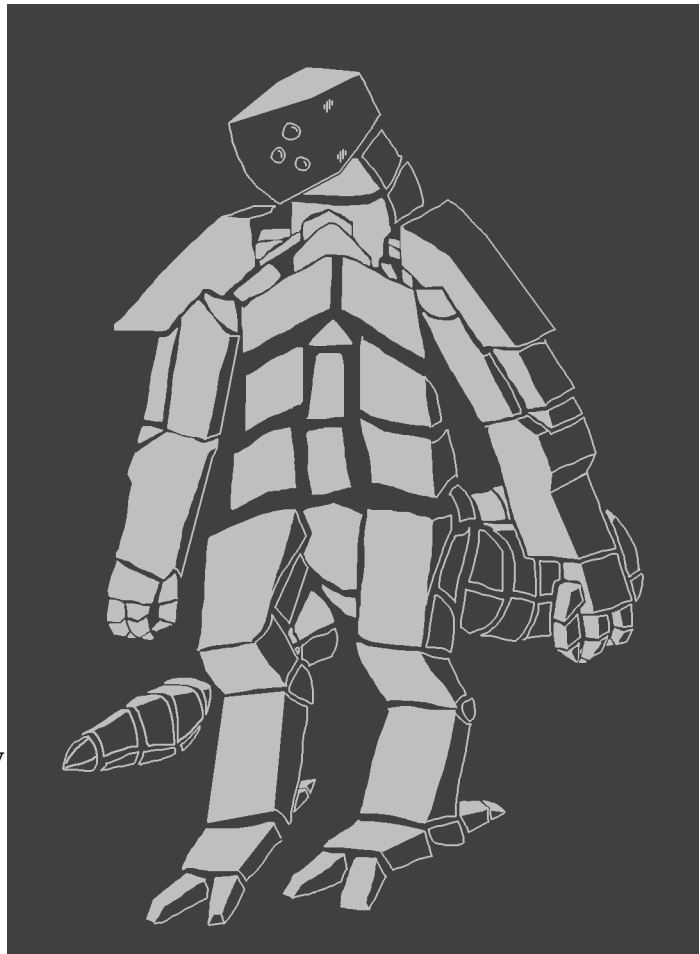
With all that shielding it *could* also resist large amounts of small arms fire, but it's obviously not meant for that use.

It's entered from the back & closed hydraulically, requiring power to open or close, though it does have a mechanical backup with control located inside the suit near the abdomen, to be operated by her flesh tendrils (see 12.3) in case of emergency.

The suit has it's own batteries, but not a power source – it just relies on being recharged from some external power source, or possibly directly from patricia's body when that's really necessary.

It's only slightly bigger than her, making her ~2.38m tall while wearing it.

For fairly obvious reasons it usually only travels with her in form of a blueprint, which given it's low-tech nature she can hire any local manufacturing company to manufacture, although other times it can be quite useful for actually aiding the travel, letting her carry a lot more than she ordinarily would be able to.



13.5 Everyday carry

1. the laptop
2. the laser phone
3. a medkit
4. a personal technical kit (need more details here)
5. a physical copy of the documentation of her artificial body
printed on 580 tough polymer pages & bound in hard cover

14 Story

14.1 Beginnings & Place of origin

Patricia was born away from earth as Sanvya, a baseline human male mostly unchanged by novel evolutionary pressures of space & only marginally touched by genetic engineering. His birthplace was a typical o'Neill cylinder, embeded inside a mined-out asteroid for protection & powered mainly by nuclear fission. It was an isolationist community, with the cyllinder keeping a distance of at least several light-seconds from any others, and made of locally produced steel, being 6km wide & 28km long & housing roughly 1.8 million people.

The habitat was first founded by a community seeking isolation to accelerate their research on artificial intelligence by not implementing safety regulations, but as all kinds of other needed workers imigrated, bringing in families & general variety into the community, the cause diluted & it ultimately became a generic place of living for people of all kinds, with the only remains of it's original purpose being a group using threats from a future superintelligent AI to coerce it's members into recruiting more talented people & working on the creation of said superintelligent AI.

14.2 Childhood & young adulthood

Sanvya was raised with values of family first & his parents didn't make much use of technologies commonly used to support the raising of a child, weary of them due to the existence of what they saw as a tech cult in the habitat they lived in. He was always quiet & seemingly struggled with social relationships even though in his own opinions he was doing perfectly fine. He'd hardly remember anyone he went to school with to the point he had to ask people he'd seen for years what their names are, save for a few friends who he'd get very close to but then fail to stay in contact with later.

Most people he considered his real friends would be people who lived in different habitats, with whom he always stayed in contact, neverminding the communication delays that would be infuriating to anyone else.

He had a very general fondness for science & a particular more focussed one on topics of transhumanism, as well as open source software. Designing a full artificial body for himself which would use only commonly available or easily manufacturable components was a lifelong project that soaked up a lot of his free time, and which he never revealed to his parents.

He continued his education past the mandatory minimum, focussing on IT & ended up getting a degree in software engineering despite finding halfway through the studies that he actually did not particularly like it, but considered it a good thing, saying that if he had actually studied what he liked it would have only made him hate it.

His relations with his parents were very tense, at least from his side, as they never really discovered many of the issues he had with them due to him being so uncommunicative. After gradually looking for work in other habitats more & more, one day he simply booked a flight out to one where he was accepted for work & simply neglected to stay in contact with his parents afterwards.

14.3 First proper job

The first job he got, not counting the internships he went through as part of his studies, was one that made any use of his skills in the IT field. The reason for his hiring was the fact of him having an engineering degree & his enthusiasm for working with numbers, though a degree in material science would be preferable, and also the descriptions of his personality, which he would describe as diplomatic & detail-oriented, with tendencies for strictly adhering to plans made ahead of time.

The job he worked was in assisting research of properties of various materials, which was almost completely automated & so did not actually require much done, meaning rarely more than 8 hours of work per week & most days completely free, but still required people for checking on the proper functioning of the machines doing the testing. His strict adherence to established procedures & the willingness to document everything made him a good fit for the job & he would provide very accurate information about the accuracy of the data provided by the machines to people who did the more scientifically demanding parts of the work & occasionally would warn them of the data produced by the machines being unuseable due to some kind of error in their function. The job paid him yearly an equivalent of about 3.5 million dollars in today's terms.

14.4 Progressive augmentation & finalizing plans

Having taken note of the new culture he found himself in being a lot more generally accepting of extensively augmented people rather than being mostly two extremes about it, he started experimenting with augmentation. First it was just a device that exposed a port on the skin of his arm, which he could connect to any device with a cable, which would emulate a mouse & keyboard using signals directly from his nerves. It took him a long time to get proficient with it, but in succeeding at using it he proved to himself that he's able to adapt to such things.

What followed was him starting a project to finalize all the documents he'd written detailing the artificial body he wants into a full schematic that could actually be manufactured with the proper documentation. Over the course of 3 years he'd talk to many engineers & doctors about various aspects of it, getting input for designing various parts of it or validating them – pretty much all of the design ended up relying on pre-existing components which had been in use for centuries & very well known, partly for the certainty that they'll work & partly from necessity, as anything fundamentally new would itself take years to validate properly. In that time he also augmented himself even more – firstly more nerve ports for controlling external electronics with nerve activity, then ones that would also let him feel things input into them as electronic signals, eventually after getting used to those an eyeball replacement which could see any frequency of light between infrared & ultraviolet.

His coworkers took note of the augmentation appearing on his body, but encouragingly did not seem bothered or very interested at all. Once the design of his new body was finalized he communicated with his employers to leave the job at a time when it would not disrupt any research work, as he would certainly need a long recovery time from his further plans & valued it quite much that his personal goals do not interfere with things he had obligated himself to.

14.5 Surgery, recovery & change of identity

Sanvya made use of his employers in finding the most adequate & trustworthy manufacturing companies as well as medical professionals to oversee his surgery. After half a year of preparations that included having his new body manufactured & tested, planning out his surgery & discussing the details with medical professionals, he had his surgery. His biological body was reduced from the full human form to a collection of the internal organs of his torso, connected together with artificial tissue & contained in a protective bag with connection ports, with several organs added, some fully artificial & some only in part, constructed from tissue which already existed elsewhere in his body, which would all then be connected into & closed inside the new artificial body.

The remains of his body that were removed would afterwards be stored in preservation for a fee by the medical facility in case they are ever needed again.

He would need to learn how to use his hands again, walk, stay upright at all, and even recognize the objects he's seeing as the new visual input was not quite the same as that of his natural eyeballs – at the very least having one artificial eye before allowed this to be mitigated by copying the connections from it.

It took pretty much the whole following year for him to become proficient enough with using his new body to be released home to continue on his own. He'd gradually enable the various features of his body over half of the following year to slowly get used to them, but found that he got quite good at getting used to new things after nearly a year of having a completely new body which required completely different inputs to move correctly.

After some legal arrangements Sanvya had their legal name changed, now their name would be Patricia, and last name changed from Foidbaser to Whateverweaver, which was actually the nickname they'd been using on the internet for interacting with most of their friends. To pretty much anyone who knew her before as Sanvya, she'd effectively disappear after that point, as her whole social life moved on with her new identity.

After the success of her transformation, she decided to release the full blueprint & documentation of her body on the internet for free, for anyone who wishes to go through a similar process.

(so from what I've written, from the start of creating the actual final design of her body to actually being in it & using it normally, had taken 5 years)

14.6 Work aboard asteroid mining vessel (todo more detail)

Patricia saved up some funds for safety, to cover any potential emergencies, and traveled out to Ceres to get herself hired by a mining crew. After a week or so of advertising around to various crews, she got picked up by one to work as the onboard engineer on their asteroid mining spacecraft. She kept things working & fixed those that weren't working. They quite liked her for keeping to herself & not causing problems, as well as for the extra Delta-V they could get by simply not having food for her onboard beyond some mass-efficient but ostensibly disgusting nutrient pastes of mostly fat & protein.

She worked for a share of the profits, as the inconsistency of mining made consistent pay effectively impossible. After several months they had a good find & the small share Patricia got was still enough to make her moderately rich. She stayed for longer, wanting to learn more from the other onboard engineers, particularly in operation of the nuclear drive of the ship, which gave her experience that she later used to get a more officially recognizable license for handling fissile material.

15 Math & reasoning

15.1 Radioisotope ThermoPhotoElectric Generator

[Thermophotovoltaics](#) are limited by [carnot efficiency](#), and the wikipedia page about [General Purpose Heat Sources](#) mentions that they can get up to 600°C. That claim has a “citation needed” attached to it but given the red glow of the shown GPHSs I figured it’s right enough.

Given that, I calculate the efficiency with this formula:

$$\eta = (T_H - T_C) \div T_H$$

Where η is the carnot efficiency

T_H is the temperature of the heat source (in this case 600°C = 873.15°K)

T_C is temperature of the collector (I assumed 40°C, so 313.15°K)

η comes out to 0.64135601, I just figured I’ll bump that down to 0.5 to not be operating at the very limits of what is possible.

Each GPHS creates 250 watts of heat, and she uses two, so 500W of heating & 250W of electricity.

I also went ahead & calculated whether that might cook her alive if she’s stuck in vacuum, having to rely on the thermal radiation of her body to get rid of the heat.

Rate at which energy is lost to thermal radiation from a given surface at a given temperature can be gotten from the [Stefan-Boltzman](#) law:

$$P = A\epsilon\sigma T^4$$

Where P is the power (in this case 500W, since that’s how much heat the RTPVG is producing)

A is the area of the radiating surface (my best estimate for the area of patricia’s body was 1.71m²)

ϵ is [emmisivity](#) (for human skin it’s around 0.99, for teflon 0.85, I figured 0.9 for patricia is ok)

σ is the Stefan-Boltzman constant (I used $5.670374419 \times 10^{-8} \text{Wm}^{-2}\text{K}^{-4}$)

T is temperature of the radiating surface (this is the unknown)

Rearranging that to get the temperature at which she radiates out as much heats as she gains from the RTPVG I get the equation: $(P \div A\epsilon\sigma)^{1/4} = T$

With the given data it returns $T=275.125^\circ\text{K}$, almost 2°C. Losing heat is a lot easier when you have an atmosphere to conduct it away to, so with that I conclude: heat produced by the RTG will not be an issue.

If she’s really spaced then heating from the sun would be a problem too, but exploring that was not the purpose of this chapter. I’ll explore that idea more in another chapter & hopefully remember to change this sentence to link to that chapter.

15.2 RCS system – pressure tank, ISP & impulse

I said she uses pressurized air to move around easier in 0g, here are the specifics.

The tank located in her tail would be a small spherical tank 6 cm in radius internally & 7cm in radius externally.

From the volume of a sphere, that gives it an internal volume of 0.000904779m³ or 0.9L

Let's say it's made of Stainless Steel, with a Yield Strength of 275MPa.

According to the [wikipedia page about Pressure Vessels](#) (Jan 1 2024), stress in the walls of a spherical pressure vessel can be calculated using this formula:

$$S = \frac{p (r + 0.2t)}{2tE}$$

Where:

S is the stress

p is pressure inside the vessel

r is the inner radius

t is thickness of the wall

E is seam efficiency, assume 1 when there is no seam

In our case, E=1, t=0.01m, r=0.06m, S=275×10⁶Pa and p is unknown.

Rearranging the equation we can get $p=(2tES)/(r+0.2t)$, from which p comes out to 88.71MPa, or 887 Bar. Given that we assumed stress to be yield strength, this is the pressure at which the vessel will burst, so it ofcourse cannot be used as the operating pressure. The vessel will instead operate at maximum 450 bar, roughly half of what it can handle.

The density of air at 1 bar is 1.2kg/m³. From the ideal gas law, increasing pressure to 450 bar would result in a 450-fold decrease in volume for the same amount of substance, if temperature remains the same. That 0.9L tank is then able to store an equivalent of 405L or 0.405m³ of air at atmospheric pressure, which translates to 0.486kg of air.

The wikipedia page on cold gas thrusters claims using nitrogen as fuel can provide an ISP of 80s in theory, and in practice 73s, while using CO₂ can provide an ISP of 67s in theory, and 61s in practice. Given the nozzles on her body would not be so much optimized for efficiency as for not being too visible, I would figure an ISP of 65s may be fitting.

ISP of X seconds means 1kg of propellant can provide kg-force of thrust for X seconds, I kinda hate the reliance on earth's gravity for the definition but at least it's clear.

With an ISP of 65s & 0.486kg of propellant, that's 1kg of force (9.81 newtons) for 31.6s. The total impulse she stores then is 310Ns.

With her body massing 133.6kg, that's 2.32m/s of potential velocity change.

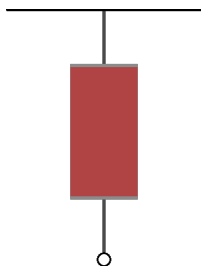
For water, the ISP is 150s & there's 0.75kg of propellant, that makes for 1103.625Ns of impulse, and with her body mass that is 8.256m/s of potential velocity change.

15.3 Muscles & strength

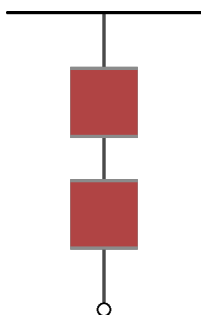
15.3.1 Trading contraction length for contraction strength

The strength of a muscle is a function of its cross-section, and the change in length upon contraction is a function of the muscle's length. After having given this some thought I found a pretty elegant way to trade distance for force without using the usual block & tackle systems which I think would cause a lot of additional internal wear if used due to friction on the rotating parts.

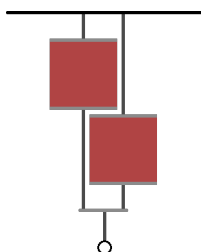
let's start with just a single artificial muscle in an imaginary lab setup:



The force with which a muscle pulls is independent of its length because any given horizontal slice of the muscle is pulling with the same force – effectively the same as if it were not actively pulling but simply transmitting the force further, so it should not matter if the muscle were split in half & the two halves connected by a nonelastic object.



Splitting the muscle like this truly changes nothing: each half is half as long so the distance it will move when it contracts is half as much, but there is two of them & one moves the other when it contracts. The cross-section of each half's is the same as before, but one pulls through the other, so the force they pull with together is actually the same as the force only one of the halves would pull with – let's rearrange them to use that:



The muscle halves are now not pulling on one another, each half exerts force directly between the mounting point & the working end, so their forces add up, but now the end is only lifted half as much. I could have put them side by side but this odd arrangement shows the benefit better: you can cleverly arrange them & possibly fit one right under the other if you do it right, so the new arrangement is taking up as much space as the original muscle we started with.

15.3.2 Mechanism of Constrained Liquid Dielectric Muscles

Dielectric Elastomers are effectively capacitors with both the conductive film & the insulator between the two separated layers of it picked to be elastic. When the two layers of film are charged relative to each other, they attract, and compress the elastic insulator.

I could not find a solid elastic insulator with low enough young's modulus to let itself be significant compressed, having tried teflon & silicone rubber, so I went with a slightly more advanced design: instead of a solid dielectric made of one material, I am using deionized water, closed in effectively very flat nonelastic balloons that aren't filled to full capacity & can be squeezed to half height before there is no more material left on the side to let the liquid flow away from the middle. Instead of compressing a solid & relying on it having some give to it, I am simply moving a liquid in a constrained space, which effectively sets young's modulus for it to 0. To further protect against them popping under their own strength they would be enclosed in a sleeve that does not let them expand horizontally beyond what it takes to contract the muscle by 50%.

Two formulas will be of use here, the equivalent electrostatic pressure formula & capacitance formula:

$$p_{eq} = \epsilon_0 \epsilon_r (U/z)^2$$

$$C = \epsilon_0 \epsilon_r (A/z)$$

Where:

P_{eq} is the equivalent electromechanical pressure, it compresses the elastomer.

C is capacity.

ϵ_0 is [vacuum permittivity](#), I take for it $8.854187812813 \times 10^{-12} \text{F/m}$

ϵ_r is [relative permittivity](#), which describes the absolute permittivity of a material in terms of how much larger it is than the previously mentioned vacuum permittivity.

U is voltage

A is area

z is separation between the two plates of a capacitor

Now to turn variables into constants based on what I already know I'll be using:

The shell containing the fluid is of negligible thickness, so I can ignore it.

I would assume 2mm separation between electrodes. (though when the muscle is contracted, that will reduce to 1mm)

The Dielectric Constant for Purified Water, is 78 based on information from [the wikipedia page "Liquid Dielectric"](#) (where I am assuming Purified means pure H_2O , no ions or minerals). With that, a lot of variables in the equations have been simplified to constants:

$$p_{eq} = 6.90626649399414 \times 10^{-10} \text{F/m} \times (U/0.002\text{m})^2$$

$$C = 6.90626649399414 \times 10^{-10} \text{F/m} (A/0.002\text{m})$$

Contracting the muscle decreases the separation between the plates, which increases the capacitance. Capacitance C describes how much charge q is needed in the capacitor to achieve a given voltage U :

$$C = q/U$$

Charge cannot change without either leaving the capacitor or more entering, so voltage must change to compensate. As the muscle contracts, voltage drops, but the pressure within the muscle is based on the fraction of voltage over separation, so the force does not change as voltage stays proportional to the separation between the plates.

The voltage is limited by the liquid dielectric used in the muscle – the voltage across the layer of the dielectric must not surpass its breakdown voltage. Breakdown voltage is linearly related to the length of the dielectric across which a voltage is applied. In this context that is the separation between the plates of the capacitor. Since voltage decreases linearly with the decrease in separation, and breakdown voltage also decreases linearly with the decrease in separation, charging the muscle to an voltage which does not trigger electrical breakdown of the dielectric in its current state is safe, as while the muscle contracts, the voltage across the plates will decrease to stay below the breakdown voltage as it also decreases.

If no losses to resistance or other factors occur, all of the electrical energy put into the muscle should be possible to recover, unless external factors act upon the muscle to stretch or compress it, changing the capacitance in a way not related to the charge provided. If the muscle is stretched externally, capacitance will decrease & consequently the voltage will increase. This can be used to turn physical work (stretching the muscle, force applied across distance) into electrical energy (removing charge from the capacitor at a higher voltage than had been used to put it in there). This also explains the expenditure of energy through the use of this muscle – when the muscle is used to do physical work, it exerts force, which means it is stretched – in that state its capacitance is lower than it normally would be, so higher voltage is needed to put the same amount of charge in it. Then once the work is done, the muscle is no longer being stretched, so when the charge is pulled back out to relax it, it is pulled out at a lower voltage than had been used to put it in the muscle.

15.3.3 Absolute theoretical limits of the muscle's operation

Now for material limitations of what the muscle can do:

The Dielectric Strength of Distilled Water, which I assume to be equal or more conductive than Deionized Water is 60MV/m (again from [the wikipedia page "Liquid Dielectric"](#)) so for the 2mm of separation the voltage cannot exceed 120 kilovolts, or else the muscle will electrically break down & short circuit.

From the earlier given simplified formula ($p_{eq} = 6.90626649399414 \times 10^{-10} \text{ F/m} \times (U \div 0.002\text{m})^2$) setting U to exactly 120 kilovolts results in $p_{eq} = 2486255.93783789\text{Pa}$ or 2.486256MPa

The strength of the muscle is 2.486256MPa if pushed to the absolute limit, which in more useable units is 248.62556N/cm².

For a 48cm bicep 5cm in diameter (19.635cm² cross-sectional area) that is a contraction of 24cm & force of 4881.76N *at the joint*. Using the trick I talked about at the start that could be turned into a 12cm contraction that carries a force of 9763.52N.

Further lever effects will reduce that, so for example patricia's arm, being a 3rd class lever & moving at the wrist 10 times more distance than the point where the force is applied, will turn that into 976.352N, equivalent to the weight of 99.5kg in 1g.

That was the absolute limit of what is possible, in reality you wouldn't want to be getting within a small twitch of the voltage that will cause electrical breakdown, maybe only going up to 90% would be better. The strength of the muscles squares with the voltage they use, so if you're using 0.9 times the maximum voltage you're getting $0.9^2=0.81$ times the maximum strength. This will make the numbers calculated here useful later.

A liquid dielectric with higher dielectric constant or higher dielectric strength would help.

15.3.4 Power consumption

Since these muscles are effectively just very flexible capacitors, I can calculate how much actual energy it would take to move them. As stated in the theoretical section, all of the energy put into the muscle can be pulled back out, but it's still possible to calculate the energy that will be put into the muscle in form of electrical charge at some voltage, & transfer of energy is almost never perfectly efficient, so we can conceptualize the losses as some fraction of that energy.

I'll again be using the example of a 48cm bicep with a 5cm diameter.

Each 2mm slice of the muscle is itself a capacitor, meaning there's 5 of them in 1cm of muscle.

The formula for capacitance of a single capacitor is as previously stated: $C = \epsilon_0 \epsilon_r (A \div z)$

I've already shortened it to this given what variables we already know the value of:

$$C = 6.90626649399414 \times 10^{-10} \text{ F/m} \times (A \div 0.002\text{m})$$

Since every 2mm slice is such a capacitor, we can generalize this equation to calculate the capacitance of the whole muscle by taking the length of the muscle into account as L:

$$C = 6.90626649399414 \times 10^{-10} \text{ F/m} \times (A \div 0.002\text{m}) \times (L \div 0.002\text{m}) = \\ 6.90626649399414 \times 10^{-10} \text{ F/m} \times (AL \div 0.000004\text{m}^2)$$

Area $\times$ Length = Volume so it could also be:

$$6.90626649399414 \times 10^{-10} \text{ F/m} \times (V \div 0.000004\text{m}^2)$$

or more simply

$$1.7265666235 \times 10^{-3} \text{ F/m}^3 \times V$$

Assuming a perfect cylinder, the volume of a 48cm bicep with a 5cm diameter is 942.477795cm^3 or 0.000942478m^3 . Plugging that into the equation results in 1.62725 microfarads

Such a capacitor, if charged up to 120 kilovolts would hold 195.27 milicoulombs of charge, which given that voltage is 23432.4 joules of energy. The energy will be reclaimed & returned to storage with some percentage of inefficiency, so the fraction of this energy that could not be reclaimed due to various losses would be the amount of energy which is lost to heat from one full contraction when operating right at breakdown voltage. This may be an arbitrary constant between 1 and 0 or 100% to 0%, eg. 1% losses would mean in this case 23.4J is lost to heat. This could also be said in terms of efficiency, where 99% efficiency means 99% of the energy cycled through the muscle is not lost to heat, meaning 1% losses.

This gives a way to calculating the losses from charging it – A coulomb is the amount of charge delivered by current of 1 ampere flowing for 1 second. Heating of a conductor through which current is passed can be described with the equation:

$$P = I^2 R$$

where P is the power, rate at which heat is added to the conductor, I is the current & R is the resistance of the conductor.

I'll use an example of a 1 metre of copper wire 2mm in diameter.

Copper has resistivity of $16.78 \times 10^{-9} \Omega \text{m}$.

The resistance of it would be $16.78 \times 10^{-9} \Omega \text{m} \times 1 \text{m} \div 12.57 \text{mm}^2 = 0.001334924 \Omega$

let's say we want the muscle fully charged within 10 milliseconds.

To deliver 195.27 milicoulombs of charge in 10 milliseconds you'd need a current of 19.527 amperes to flow for those 10 milliseconds.

Plugging that into the previously presented equation we get:

$$(19.527 \text{A})^2 \times 0.001334924 \Omega = 0.509011499 \text{W}$$

This heating happens over the 10 milliseconds for which the current flows, so in the end 0.005090115 joules of heat are produced

That is 0.000000217 times the energy which had been moved, so the losses from resistive heating here are 0.0000217%, or in other words the transfer is 99.9999783% efficient. In reality most times the losses will be twice as big, since the electricity needs to flow through the wire again when the muscle is relaxed.

As a rule of thumb using a wire X times longer would also make the losses X times bigger.

This part could use some more research, since resistive losses surely aren't the only source of inefficiency in an electric system. For now I will just treat it as such though, as I don't actually have exact knowledge of what other than this causes such losses or how much.

As I described in the 2nd part of this chapter, I have a conceptual mechanism for how doing physical work would make the muscle lose energy, but I don't quite know how to put numbers on it yet.

15.3.5 Simplified equations & standard muscle module.

The equations solved earlier to know the limit of this exact muscle's operation actually allow greatly simplifying the equations using the two discovered pieces of information: that pressure inside the muscle grows with the square of voltage & energy use grows linearly with the voltage.

These can be used when there is a known reference point – it does not have to be the maximum achievable value but I consider that an easy way to keep track of what should or should not be done with the muscle.

Voltage U needed to exert some force F : $U = (F/F_{\max})^{1/2} \times U_{\max}$

Energy E held within the muscle when it is charged to some voltage U : $E = (U/U_{\max})^2 \times E_{\max}$

Energy E held within the muscle when it is exerting some force F : $E = (F/F_{\max}) \times E_{\max}$

This does not really have any new physical implications, but it does mean that creating one standardized muscle module which is then used all over the body connected however many times it's needed in series or parallel would make any calculations much easier, as they could be referenced to the $U_{\max}$, $F_{\max}$ & $E_{\max}$ values calculated for that specific length & thickness of muscle.

A module that is 12cm long would contract by 6cm, and if it has a cross-section of 5cm² it would be 2.523cm in diameter.

As calculated in 15.3.3, the force of the muscle at the absolute maximum voltage is 248.62556N/cm², which means this muscle module would be exerting 1243.1278N.

The volume of a cylinder 12cm tall with a 5cm² base is 60cm³ = 0.00006m³, which can be plugged into the equation for capacitance from chapter 15.3.4: $C = 1.7265666235 \times 10^{-3} \text{F/m}^3 \times V$

$1.7265666235 \times 10^{-3} \text{F/m}^3 \times 0.00006 \text{m}^3 = 103.5 \text{ nanofarads}$

At the maximum voltage of 120kV, it would store 12.42 milicoulombs of charge.

$103.5 \times 10^{-9} \text{F} \times 120 \times 10^3 \text{V} = 0.01242 \text{C}$

Which at that voltage would contain 1490.4 joules of energy.

$12.42 \times 10^{-3} \times 120 \times 10^3 \text{V} = 1490.4 \text{J}$

So these are the characteristics of a single standard muscle module:

12cm total length, 6cm contraction length D. ($D_{\text{std}} = 6\text{cm}$)

Volume V_{std} : 60cm³ (0.00006m³)

Max force F_{std} : 1243.1278N

Max stored energy E_{std} : 1490.4J

These numbers may be interchangeable with the corresponding physical quantity with “std” subscript, short for standard. Muscles not marked otherwise would have these characteristics.

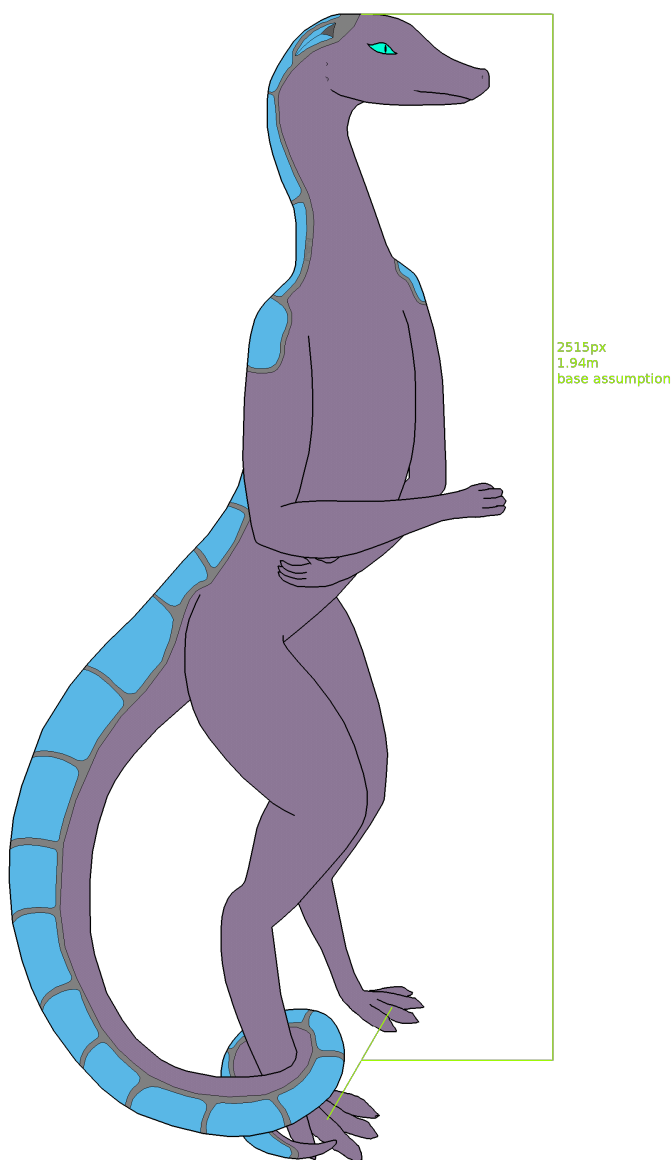
Put in parallel, modules add together their forces. In series, they add together contraction lengths.

15.3.6 Patricia's actual strength

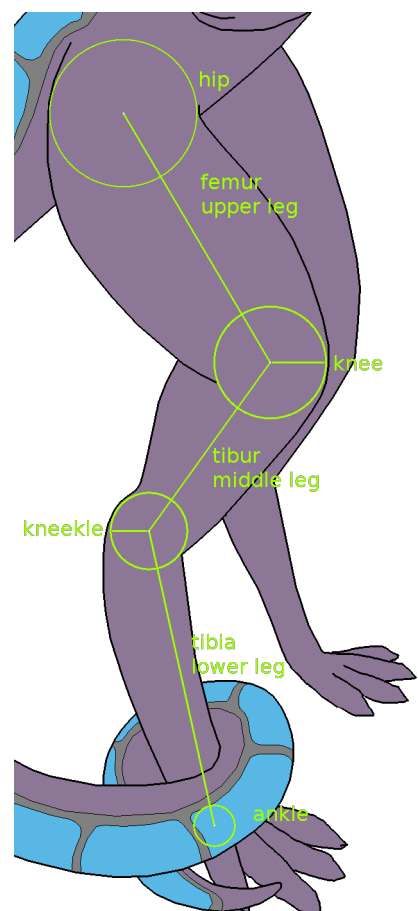
1. measuring patricia (and establishing vocabulary)

For anyone concerned that I am measuring a drawing instead of making drawings based on numbers I calculated beforehand, I do actually have quite a bit of experience trying both, and I've just found that coming up with numbers beforehand results in things that work in theory but are simply impossible to make look right when actually implemented in a drawing, or sometimes simply impossible to make look like anything at all. The process of drawing a character clears up a lot of anatomical issues in a way that is easy & in my experience more accurate than mathematics. Yet better results could come from a back & forth process of drawing a character, measuring what is right or wrong about them, redrawing them to fix these problems, measuring again & so on, but at least for now that kind of effort is beyond me.

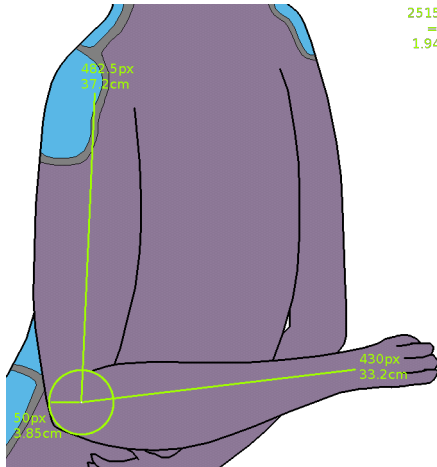
The drawing below shall serve all pixel measurements - 2515 pixels correspond to 1.94 meters.



And to be clear on the vocabulary in her odd leg anatomy, since they are not actually digitigrade, rather just have one additional bone & joint, this is how I refer to all the bones, joints & sections of the leg



2. Arm



2515px = 1.94m Patricia's upper arm is 37.2cm long, lower arm is 33.2cm long.

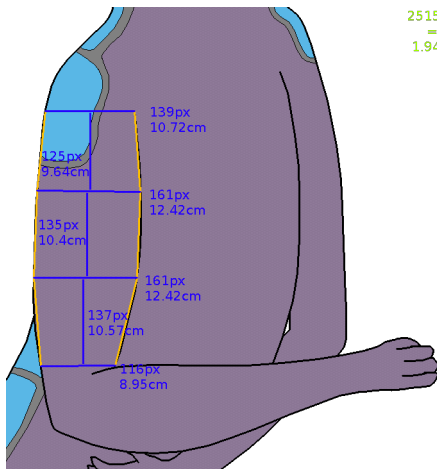
The radius of the joint at her elbow is 3.85cm, which is 8.62 times shorter than the lower arm, so forces applied there will result in 8.62 times less force & that times more movement at the wrist.

Her elbow needs to have a fairly normal human range, bending from 180° to 30°, which means her wrist would travel 150° on the circumference of a circle with the radius of it's length.

$33.2\text{cm} \times 2\pi \times (150 \div 360) = 86.92\text{cm}$
at the joint, that is 10cm of movement.

Since that's 1.19 times less than how much 2 muscle modules connected in series can contract, a 1.19 times smaller force-distance trade factor would be better, but the joint cannot be made bigger, so instead I will reduce the movement required to 6cm.

If the joint is supposed to convert 6cm of movement at itself into 86.92cm at the wrist, it's radius must be 14.487 times less than the length of the lower arm – 2.292cm. Then all 6cm of the contraction of a muscle module are used. In this case all muscle modules will be connected in paralel, with their forces adding up.



2515px = 1.94m To get the volume useable for muscle, I will model patricia's arm as 2 frustums & 1 cyllinder, with their volumes added & multiplied by 2/3 as I imagine the arm being a little thinner in one dimension rather than having a perfectly circular cross-section. I also decrease radius by 1cm to account for skin.

To get the volume of the frustums I use this formula:

$$V = (1/3) \times \pi \times h \times (r^2 + r \times R + R^2)$$

To get the volume of the cyllinder I use this formula:

$$V = h \times \pi \times r^2$$

$$V_1 = (1/3) \times \pi \times 9.64\text{cm} \times ((5.36\text{cm}-1\text{cm})^2 + (5.36\text{cm}-1\text{cm}) \times (6.21\text{cm}-1\text{cm}) + (6.21\text{cm}-1\text{cm})^2) = (1/3) \times \pi \times 9.64\text{cm} \times 68.8693\text{cm}^2 = 695.234508694\text{cm}^3$$

$$V_2 = 10.4\text{cm} \times \pi \times (6.21\text{cm}-1\text{cm})^2 = 886.867333542\text{cm}^3$$

$$V_3 = (1/3) \times \pi \times 10.57\text{cm} \times ((8.95\text{cm}-1\text{cm})^2 + (8.95\text{cm}-1\text{cm}) \times (6.21\text{cm}-1\text{cm}) + (6.21\text{cm}-1\text{cm})^2) = (1/3) \times \pi \times 9.64\text{cm} \times 131.7661\text{cm}^2 = 1330.176723097\text{cm}^3$$

$$V_{\text{arm}} = (V_1 + V_2 + V_3) \times (2/3) = 2912.278565333\text{cm}^3 \times (2/3) = 1941.519043555\text{cm}^3$$

I will give 40% of that volume each to the bicep & tricep, while leaving 20% for the bone & other internals of the arm. That would mean the bicep gets $0.4 \times V_{\text{arm}} = 776.6 \text{cm}^3$. This is enough for 12.94 standard muscle modules. Using whole 13 would take up 780cm^3 which bumps up the use of total volume to 40.2% - that is acceptable, the bone will have to fit in the remaining 19.6% of the arm's volume then.

13 standard muscle modules put in paralel can exert force of $13 \times F_{\text{std}}$ which the elbow joint trades for movement distance at a factor of 14.487, which means the force at her wrist is $0.89735625 \times F_{\text{std}}$

$$0.89735625 \times F_{\text{std}} = 1115.5 \text{N}$$

Equivalent to the weight of 113.7kg in 1g.

In reality though, she never would push the voltage above 90%, meaning the real force will never reach more than 81% of that.

The actual force she can exert at the wrist then is 903.5N, equivalent to the weight of 92kg in 1g.

Tricep is identical to the bicep.

If you're confused by me sticking to F_{std} so hard here, I am doing this so that if I ever make a modifiation to the standard muscle module which only changes the value of F_{std} I will not have to redo all the math, and instead only will have to convert the results measured in F_{std} to results in newtons & g-kilograms.

3. Hand

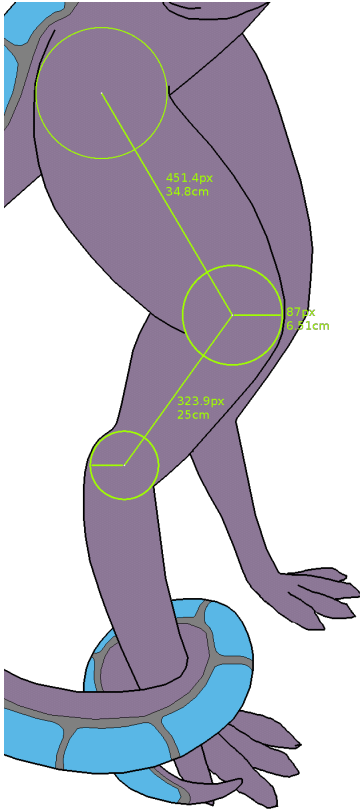
todo

for now I'm making a completely thoughtless assumption that the bending of each of her fingers is controlled by 2 muscle modules (one for flexion, other for extension), two more for each finger to control the spread (again, positioned so one does the reverse of the other) & 6 to control the bending & turning of the wrist, making for a total of 26 muscle modules with collective volume of 1560cm^3 .

This would mean each of the fingers can exert a force of 1243N , which is somewhere around 10 times more than any normal human could do.

Ps: this seemed a lot more reasonable till I fixed my math to actually use radiuses instead of diameters where radiuses belonged

4. Leg



Patricia's tibia is 25cm long & her knee is 6.51cm in radius, meaning movement of the joint translates to 3.84 times bigger movement of the kneelke.

Her knee should have a range of movement between 20° & 180°, which means that her kneelke would be able to move 69.81cm.

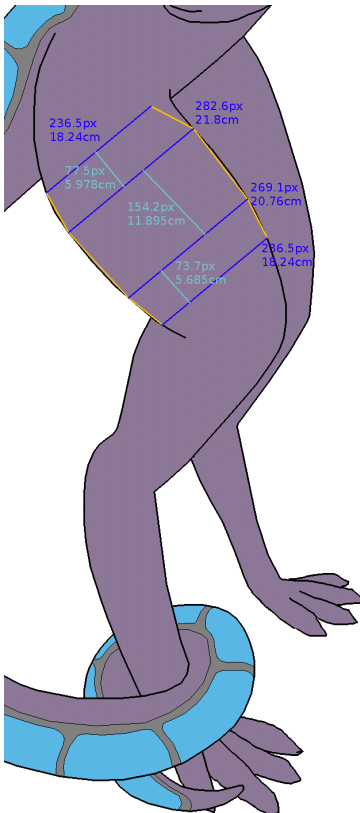
$$25\text{cm} \times 2\pi \times ((180-20) \div 360) = 69.81317\text{cm}$$

That requires the joint to move 18.18cm, which is a bit more than 3 standard muscle modules in series can contract by. Unfortunately 3 muscle modules in series would be 36cm long, but her thigh is only 34.8cm long, which means at most 2 modules in series can be used.

2 modules in series contract by 12cm.

For 12cm of movement to translate into 69.81cm of movement, it needs to be increased by a factor of 5.82, so the radius of the knee must be 5.82 times smaller than the tibia, which is 4.296cm.

In this case, muscle modules are installed in pairs of 2 modules in series & the force they exert results in a force 5.82 times smaller at the kneelke.



I will approximate the volume of patricia's thigh with 3 frustums. Their volumes will be added & multiplied by 2/3 as I imagine her thigh being slightly thinner in one direction rather than having a perfectly circular cross-section. I also will decrease the radiuses by 1cm to account for the skin.

The drawn measurements may appear to have a slight inaccuracy, but I was doing my best to consider areas that reasonably could have muscle, which may not be the entire thigh.

$$V_1 = (1/3) \times \pi \times 5.978\text{cm} \times ((9.12\text{cm}-1\text{cm})^2 + (9.12\text{cm}-1\text{cm})(10.9\text{cm}-1\text{cm}) + (10.9\text{cm}-1\text{cm})^2) = (1/3) \times \pi \times 5.978\text{cm} \times 244.3324\text{cm}^2 = 1529.556731\text{cm}^3$$

$$V_2 = (1/3) \times \pi \times 11.895\text{cm} \times ((10.38\text{cm}-1\text{cm})^2 + (10.38\text{cm}-1\text{cm})(10.9\text{cm}-1\text{cm}) + (10.9\text{cm}-1\text{cm})^2) = (1/3) \times \pi \times 11.895\text{cm} \times 278.8564\text{cm}^2 = 3316.996878\text{cm}^3$$

$$V_3 = (1/3) \times \pi \times 5.685\text{cm} \times ((10.38\text{cm}-1\text{cm})^2 + (10.38\text{cm}-1\text{cm})(9.12\text{cm}-1\text{cm}) + (9.12\text{cm}-1\text{cm})^2) = (1/3) \times \pi \times 5.685\text{cm} \times 256.7044\text{cm}^2 = 1459.364514\text{cm}^3$$

$$V_{\text{thigh}} = (V_1 + V_2 + V_3) \times (2/3) = 6305.918123\text{cm}^3 \times (2/3) = 4203.945415333\text{cm}^3$$

I think it reasonable to do the same as with the arm, giving the 2 sets of muscles each 40% of the volume of the thigh & leaving 20% for the bone & other internals.

$$0.4 \times V_{\text{thigh}} = 0.4 \times 4203.945415333 \text{cm}^3 = 1681.578166133 \text{cm}^3 = 28.026302769 \times V_{\text{std}}$$

28 muscle modules may be used, and since they may be installed in pairs of 2 modules in parallel, there will be 14 such pairs.

As 2 muscle modules connected in series exert the same force as 1 module, the whole bunch exerts $14F_{\text{std}}$

The knee trades force for travel distance at a factor of 5.82, so the force at the knee is that many times smaller: $2.405498282 \times F_{\text{std}}$

$$2.405498282 \times F_{\text{std}} = 2.405498282 \times 1243.1278 \text{N} = 2990 \text{N}$$

This force is equivalent to 304.8kg in 1g

In reality though, she never would push the voltage above 90%, meaning the real force will never reach more than 81% of that.

The actual force then can at maximum be 2422N, equivalent to 246.9kg in 1g

This would mean she can still walk while carrying 113.3kg on top of her own mass in 1g.

This alone is actually enough to calculate max jump velocity & max jump height under 1g using [equations for a falling body](#). Note here that she has 2 legs, so the force is doubled, but also her weight is subtracted from the net force.

$$F_{\text{net}} = (2 \times 2422 \text{N}) - (133.6 \text{kg} \times 9.81 \text{m/s}^2) = 3533 \text{N}$$

She masses 133.6kg, so we can calculate her acceleration:

$$a = 3533 \text{N} \div 133.6 \text{kg} = 26.445 \text{m/s}^2 \text{ (2.696g)}$$

a body under acceleration a takes $t = \sqrt{2d \div a}$ to travel distance d.

$$t = \sqrt{2 \times 1 \text{m} \div 26.445 \text{m/s}^2} = 0.275 \text{s}$$

$$v = 26.445 \text{m/s}^2 \times 0.275 \text{s} = 7.27 \text{m/s}$$

Again with the equations for a falling body, distance d travelled by an object falling for time t with acceleration g is:

$$d = (1/2) \times g t^2$$

& time t taken for an object to accelerate to velocity v under acceleration g is:

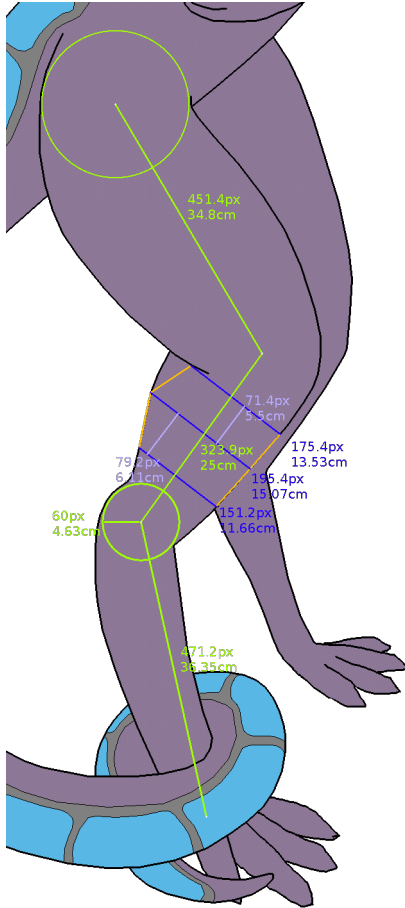
$$t = v/g$$

which in this case:

$$t = 7.27 \text{m/s} \div 9.81 \text{m/s}^2 = 0.741 \text{s}$$

$$d = (1/2) \times 9.81 \text{m/s}^2 \times (0.741 \text{s})^2 = 2.693 \text{m}$$

MIDDLE LEG – the limiter on movement speed



just like with the thigh you might see I did not quite match the shape but here it's because I actually can't quite tell what I drew to be honest. It looked fine until I tried to measure it & found that it does not really come together as a 3d shape, so I approximated something more reasonable instead.

It's volume will, just like before, be approximated by 2 truncated cones, with radii shrunk by 1cm to account for the skin, and multiplied by 2/3 to account for being a bit thinner on one axis.

$$V_1 = (1/3) \times \pi \times 5.5\text{cm} \times (5.765^2\text{cm}^2 + (5.765\text{cm} \times 6.535\text{cm}) + 6.535^2\text{cm}^2) = 654.379599486\text{cm}^3$$

$$V_2 = (1/3) \times \pi \times 6.11\text{cm} \times (4.83^2\text{cm}^2 + (4.83\text{cm} \times 6.535\text{cm}) + 6.535^2\text{cm}^2) = 624.476320229\text{cm}^3$$

$$V = (2/3) \times (V_1 + V_2) = 852.57\text{cm}^3 = 14.21V_{\text{std}}$$

Just like before, the volume is split 40%-40%-20% for the muscle extending the joint, contracting the joint, and the volume for bones & other things.

$$V_{\text{muscle}} = 5.68V_{\text{std}}$$

The muscle then has 5 standard muscle modules

The kneekle should have a movement range of 160° just like the knee. As the tibia is 36.35cm long that means the ankle would move 101.51cm on an arc around the kneekle joint.

$$2\pi \times 36.35\text{cm} \times (160^\circ / 360^\circ) = 101.508349296\text{cm}$$

A standard muscle module contracts by 6cm when activated so for 6cm of pull to translate into 101.51cm of movement the joint has to multiply the movement by a factor of 16.9, which means it's radius is that many times smaller than the length of the tibia – 2.15cm

$$5F_{\text{std}} \div 16.9 = 0.295857988 F_{\text{std}}$$

As the voltage in the muscle is kept to 90% of maximum it limits the force to 81% of maximum

$$0.295857988 F_{\text{std}} \times 0.81 = 0.23964497 F_{\text{std}} = 0.23964497 \times 1243.1278\text{N} = 297.9\text{N}$$

This is the force she can apply by her ankle. Equivalent to 30.37kg in 1g

Now for movement speed, I will have to make some assumptions

Assuming the leg there to be just a cylinder with diameter 8.5cm & length 36.35cm it has a volume of 2062.68cm^3 which, if to be given the same density as water (by design to feel like human flesh) it would mass 2.063kg

Knowing no better (because I don't know how to do any more advanced math than that) I will just assume that's a point mass located at the ankle.

297.9N acting on 2.063kg accelerates it at 144.4m/s^2

Now I can calculate the maximum speed she can run at.

If one step she takes is 1m long (that's a second assumption here) the foot will at most accelerate for half that distance before it makes contact with the ground. After that it would have to decelerate.

a body, in this case the leg abstracted as a point mass, takes time $t = \sqrt{2d \div a}$ to travel distance d when it constantly moves with acceleration a.

$$t = \sqrt{2 \times 0.5\text{m} \div 144.4\text{m/s}^2} = 0.083217833\text{s}$$

Since it's accelerating for only that long, it will reach a speed v of $v = at$

$$v = 144.4\text{m/s}^2 \times 0.083217833\text{s} = 12\text{m/s}$$

That would be the maximum speed she can run at. Any faster than that & she just cannot accelerate her foot fast enough to match the speed with which she moves relative to the ground.

12m/s is roughly 43.26km/h per hour or 26.88mph, which is a bit more than the legal speed limit of 25mph in residential areas in most of the united states of america

Now for how fast she can accelerate, I will just assume that only one leg is making contact with the ground & treat the force as applied to her whole body mass of 133.6kg, that's simply 2.23m/s^2

She would in that case take roughly 5.38s to reach her top speed.

4. walking (to be reconsidered)

I'll give this more thought after a while, for now this is gonna be a pretty simplistic calculation efficiency is the same, assuming 12m of 2mm copper diameter wire between the muscles & her power source, so 0.0002604% losses

muscles basically the same as the calculated bicep, for her leg, yes

patricia's mass is 133.6kg, which in 1g translates to 1310.6N at the ankle, so doing the lever analogy again, factor of 10 more, 13106 at the joint

for a single step $E = (13106\text{N}/4882\text{N}) \times 23432.4\text{J} = 62905.6\text{J}$

don't worry about the force being bigger than what a single muscle can supply, it's no different from just calculating it as partial contraction of multiple muscles.

0.0002604% of E is 0.164J

Practically nothing, less than a watt for any reasonable walking speed.

I will have to figure out exactly how to put numbers on that mechanism of losing energy to physical work I conceptually figured out, because it will likely be the biggest source of energy loss. For now I'm just assuming she also expends as much energy as it takes to increase her potential kinetic energy (works the same in reverse, so she gains energy from walking downhill).

15.4 Skeleton

15.4.1 Material properties

Patricia's bones will be made of material in short called [Ti-6Al-4V](#), which is an alloy which contains 90% Titanium, 6% Aluminium & 4% Vanadium.

It's properties are as such:

Yield strength	880MPa
Tensile strength	900MPa
Density	4.5g/cm ³

15.4.2 Calculation methods: 2 cases of a solid pole holding a weight

Case 1 - compressive stress: a titanium alloy pole 1m tall in vertical position, standing on an immovable object, and having a 1 ton weight placed on top of it.

If the stress is purely compressive, it can be calculated with this formula:

$$S = F \div A$$

where:

S – stress

F – force compressing the pole

A – cross-sectional area of the pole

The alloy's yield strength is 880MPa, so that is the value stress cannot exceed.

Rearranging the formula to get the Area needed such that a known force results in a given stress:

$$A = F \div S$$

$$\text{Weight of 1 ton is } 1000\text{kg} \times 9.81\text{m/s}^2 = 9810\text{N}$$

$$A = 9810\text{N} \div (880 \times 10^6\text{N/m}^2) = 0.000011148\text{m}^2 = 11.148\text{mm}^2$$

the radius r of a circle with given area A is given by this formula:

$$r = \sqrt{A \div \pi}$$

For $A=11.148\text{mm}^2$, the equation returns 1.883751207mm, which would be the absolute limit of how small the radius of such pole can be before it is not strong enough to hold up the 1 ton weight.

In reality it would be difficult to achieve purely compressive forces, and if it's somehow certain there will only appear compressive forces, then calculations should be done using either assumed bigger forces or aiming for smaller stress under intended forces, so that the stress appearing under normal operation does not reach the breaking point of whatever is being made.

Case 2 - bending stress: a titanium alloy pole 1m in length, in horizontal position, by one end attached to an immovable object & having a 1 ton weight hanging from the other end.

For stress in a beam resisting a force applied on it's end paralel to it's length, the formula is this:

$$S = M \times c \div I$$

where:

S – stress

M – bending moment applied to the beam

c – maximum distance from point on the surface of the beam from it's neutral axis

I – second moment of area of the pole

For a circular cross-section, the second moment of area is calculated as $I = \pi r^4 \div 4$

As the pole has a circular cross-section, meaning it's symetrical & it's neutral axis goes right through the center of it's cross-section, c will simply be equal to r

In that case to depend on r I replace c and I in the formula:

$$S = M \times c \div I = M \times r \div (\pi r^4 \div 4) = 4M \div \pi r^3$$

The weight of 1 ton, 9810N, is applied at a distance of 1m from the mounting point, it applies a bending moment of 9810Nm

The yeild strength of the titanium alloy is 880MPa so to get the minimum radius the stress is set to that, the only variable left is then r

To get r I rearrange the equation:

$$r = (4M \div \pi S)^{1/3}$$

$$r = (4 \times 9810 \text{Nm} \div \pi \times 880 \times 10^6 \text{Pa})^{1/3} = 0.024212083 \text{m} = 2.42 \text{cm}$$

it would be nice to get a different moment of area, taking from the wikipedia page "[List of second moments of area](#)" second moment of area for an annulus is calculated as $I = (\pi \div 4)(r_2^4 - r_1^4)$ where r_1 and r_2 are the inner & outer radius of the pipe. The equation for stress is then:

$$S = M \times c \div (\pi \div 4)(r_2^4 - r_1^4)$$

Since the shape is still circular as far as the outside surface is concerned, c will just be r_2

$$S = M \times r_2 \div (\pi \div 4)(r_2^4 - r_1^4)$$

The equation can then be rearranged to solve for r_1 :

$$(r_2^4 - (M \times r_2) \div S(\pi \div 4))^{1/4} = r_1$$

I will assume the outer radius to be 3cm & solve for the inner hollow radius using the same data:

$$r_1 = (r_2^4 - (M \times r_2) \div S(\pi \div 4))^{1/4} = (0.03^4 \text{m}^4 - (9810 \text{Nm} \times 0.03 \text{m}) \div (880 \times 10^6 \text{Pa} \times \pi \div 4))^{1/4} = 0.024896368 \text{m} = 2.489 \text{cm}$$

$$\& \text{ just to make sure: } S = M \times r_2 \div (\pi \div 4)(r_2^4 - r_1^4) = 9810 \text{Nm} \times 0.03 \text{m} \div ((\pi \div 4) \times ((0.03^4 \text{m}^4) - (0.02489^4 \text{m}^4))) = 879 \text{MPa}$$

a hollow pipe made of the alloy with radius 3cm & wall thickness 0.511cm does hold up.

15.4.3 Most stressed bone as template for all bones



As previously calculated in chapter 15.3.6 “Patricia’s actual strength”, Patricia’s each leg can exert maximum force of 2422N at the kneekle, which if that were the case would bend the tibur. I will assume a safety margin of 2 times, so the bones should be able to handle twice the force her own movements could ever exert.

I will calculate for the bone that handles the biggest forces in the entire body, so if I can fit that thickness everywhere else that’s just what all her bones will be. Her wrist is the thinnest part of any limb: 4cm in diameter, so 2cm in radius.

I will here use the previously arranged formula for the hollow radius, complete with the assumption of 3cm diameter of the bone.

$$r = (5.0625\text{cm}^4 - ((M \times 1.5\text{cm}) \div S(\pi \div 4)))^{1/4}$$

The length of her tibia is 25cm. Bending force of 2422N at 25cm applies a moment of 605.5Nm.

Yield strength of the used material is 880MPa. I want a 2 times safety margin, so the stress that should appear in the bones given these circumstances should only be 440MPa.

$$r = (5.0625 \times 10^{-8}\text{m}^4 - ((605.5\text{Nm} \times 0.015\text{m}) \div 440\text{MPa}(\pi \div 4)))^{1/4} = 0.01249086\text{m}$$

Her tibia, then, is roughly just a nitinol pole with length of 25cm & diameter of 3cm, which has a hollow cavity inside 2.5 cm in diameter.

It’s cross-sectional area is then $A = \pi 1.5^2\text{cm}^2 - \pi 1.25^2\text{cm}^2 = 2.159844949\text{cm}^2$

It’s volume would then be 53.996123725cm^3 , which with the alloy’s density of 4.5g/cm^3 means it would mass 243 grams

For now I leave it at this

As all other bones are the same, except spine, I add up the lengths of **all** her limb bones & assume they are all this thick, as well as the spine which I assume just has twice the cross-sectional area so it’s equivalent to having twice the length (yes I measured the spine, just haven’t shown that – it’s 4200px=324cm long)

$$2 \times (36.35\text{cm} + 25\text{cm} + 34.8\text{cm}) + 2 \times (37.2\text{cm} + 33.2\text{cm}) + 2 \times 324\text{cm} = 981.1\text{cm}$$

$$981.1\text{cm} \times 2.159844949\text{cm}^2 = 2119\text{cm}^3$$

$$2119\text{cm}^3 \times 4.5\text{g/cm} = 9.54\text{kg}$$

seems alright actually, but I’ve yet to consider her ribs. Maybe those would make sense to use nitinol for, you’d want them to never get permanently bent inwards.

15.5 “glowing red hot” in near-infrared night vision (WIP)

There appear to be absolutely no good resources for how hot something has to be to visibly glow as seen by a night vision camera, as it appears most of the internet does not distinguish between night vision & thermal vision – to be fair, thermal vision too is often used for night operations, to provide ability to see in otherwise complete darkness.

This chapter will be a much different than the other chapters in this section, which were mostly mathematical. I actually own a night vision camera, and so I will test this experimentally & make attempts at extracting some kind of clear result however I can. I had previously attempted to extract this information from Wien’s displacement law, but my understanding of it does not appear sufficient.

Firstly I noted that objects that to my eyes entirely ceased glowing visibly as they cooled down, still appeared white hot on the night vision camera for a while longer. This means that the temperature at which the faintest red glow can be observed is higher than the temperature at which the object glows very clearly & brightly in near infrared.

The wikipedia page about [Red Heat](#) - using colours to determine the temperature of a piece of metal – claims 525°C to be the temperature at which the red glow is “just visible”. This would mean the temperature at which a metal object will glow clearly

An addition conclusion from my experimentation is that flames appear about twice as tall on night vision as they do in the visible light.

15.6 Hypoxic brain damage mitigation by cooling & glutamate receptor blockers (todo)